

Department of Computer Science and Engineering

REPORT

System Analysis and Design of Daraz Bangladesh

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Chapter 1
Recognition of Need

1.1 Introduction

A system is a group of connected parts that work together to reach a common goal. In system analysis and design, we study how an existing system works, find out what problems it has, and then suggest ways to make it better. This process helps a business run more smoothly, serve its customers better, and grow in a stronger way. In this report, we study Daraz Bangladesh, one of the biggest online shopping platforms in the country, to find its main problems and suggest an improved system.

Daraz Bangladesh is an online marketplace where people can buy almost anything from one place. This includes electronics, clothes, shoes, groceries, home appliances, beauty products, and many other items. The platform is part of the Daraz Group, which works in several South Asian countries including Bangladesh, Pakistan, Sri Lanka, Nepal, and Myanmar. Daraz Group itself is owned by the Alibaba Group, one of the largest e-commerce companies in the world. Alibaba became the full owner of Daraz Group in 2018, which gave Daraz access to Alibaba's technology, logistics knowledge, and online payment systems.

Daraz started its operation in Bangladesh in 2015. Since then, it has grown into the country’s largest online shopping platform. Unlike a normal shop that sells its own products, Daraz mainly works as a marketplace. This means Daraz connects thousands of small and large sellers with millions of buyers across the country. A seller can open a shop on Daraz, list their products, and sell directly to customers. Daraz also runs its own "LazMall" style verified-brand section, where big and trusted brands sell their original products, giving customers extra confidence that the item is genuine. Along with the marketplace model, Daraz also manages the full order journey, which includes listing products, accepting orders, collecting payments, arranging delivery, and handling returns or refunds if a customer is not happy with a product.

Mission and Vision

Daraz Bangladesh's stated vision is to become the number one online marketplace in the country by offering good quality products and services to both sellers and customers. Its mission is built around giving customers an enjoyable shopping experience while helping sellers grow their business at the same time. To do this, Daraz says it focuses on building a large and loyal customer base, offering a wide range of products at competitive prices, and providing reliable delivery so that customers keep coming back. For sellers, Daraz aims to provide tools and

support that help small businesses reach a much bigger group of buyers than they could reach on their own.

Business Departments and Organizational Structure

Like most large e-commerce companies, Daraz Bangladesh is organized into several major departments that work together to keep the platform running. The main departments include the Seller Center, which supports sellers with onboarding, product listing, and pricing tools; the Customer Relationship Management (CRM) team, which manages communication with buyers and handles complaints; the Supply Chain Management (SCM) team, which manages warehousing, inventory, and delivery planning; the Marketing department, which runs advertising and large sale campaigns; and the Technology or IT department, which builds and maintains the website, mobile app, and backend systems. At the top of the organization sits the Managing Director, who is responsible for the overall direction of Daraz Bangladesh and who also reports to the regional Daraz Group leadership. Below the Managing Director, department heads such as the Chief Operating Officer (COO) and Chief Commercial Officer look after operations and business growth. Under these department heads, category managers, warehouse supervisors, delivery coordinators, and customer support staff carry out the daily work of the company. A simplified view of this structure is shown below.

Level	**Roles**
Top / Executive Level	Managing Director, Chief Operating Officer, Chief Commercial Officer
Middle / Management Level	Category Managers, Marketing Managers, Customer Support Managers,
Operational Level	Seller Support Staff, Warehouse Staff, Delivery Coordinators, Customer Care Representatives, Riders

Table 1.1: Simplified organizational structure of Daraz Bangladesh

Products, Payment, and Sale Campaigns

Daraz offers more than one way to pay for an order. Customers can pay online using cards, mobile financial services such as bKash or Nagad, or they can choose Cash on Delivery, which means paying the delivery person when the order arrives. Cash on Delivery is very popular in

Bangladesh because many customers still do not fully trust online payment or do not have easy access to it. Daraz also runs large sale campaigns during the year, such as "11.11" and "12.12", where products are sold at a big discount. These campaigns bring in a very large number of orders in a short time, which puts extra pressure on the whole system, from the website to the warehouse to the delivery network.

Logistics and Delivery Network

Delivery is one of the most important parts of Daraz's business, since it is the final step a customer sees before deciding whether they are happy with their shopping experience. In the early years of e-commerce in Bangladesh, delivery used to take a long time, sometimes seven to nine days for a single order. Over the years, Daraz built its own logistics network and also partnered with courier companies such as Paperfly, REDX, and eCourier to speed things up. Because of these changes, average delivery time in many areas has come down to about two to three days. Daraz also uses a central Order Management System (OMS) that helps its operations and customer service teams manage inventory, track orders, handle returns, and coordinate between warehouses, sellers, and delivery partners.

Even with all these systems in place, delivering to a whole country is not simple. Bangladesh has crowded cities, difficult traffic, and many remote areas that are harder to reach quickly. Because of this, delivery problems still happen from time to time, even though the overall network has improved a lot compared to the early days of the company.

Market Position

Daraz Bangladesh currently faces very little competition from big international companies like Amazon or eBay, since neither company runs a full local operation in the country. This has helped Daraz remain the market leader for several years, and today it is generally seen as the largest and most well-known online shopping platform in Bangladesh. However, being the market leader does not mean the system is free of problems. Because so many people use the platform every day, even small issues can affect a very large number of customers. Over time, a large number of customers have shared their experiences, both good and bad, on independent review websites, social media pages, and in local newspapers. A careful look at this feedback shows some repeating problems in the way the system works, especially around product quality, refunds, account security, delivery, and customer support.

This is exactly why Daraz Bangladesh is a good real-life example for a System Analysis and Design study. It is an active, large, and well-known company. It has a clear and repeatable business process, starting from a customer browsing the app, placing an order, making a payment, and finally receiving the product or asking for a return. Because Daraz is such a big and well-covered company, a large amount of information about it is already available in the public space, such as news articles, company statements, and customer reviews. This makes it possible to study the system closely and suggest real improvements, without needing to visit the company in person.

1.2 Problem Identification

After looking closely at customer reviews from websites such as Trustpilot, SmartCustomer, Google Play Store, and Reviews.io, and also going through local news reports, we found several problems that repeat again and again in the Daraz Bangladesh system. These problems affect how customers feel about the platform, and they also create extra work and cost for the company itself. Identifying these problems clearly is the first and most important step of system analysis, because it is not possible to design a better system without first understanding exactly what is going wrong. Below, we describe five main problems in detail, along with why each one matters and how it affects both customers and the company.

Problem 01: Product Authenticity and Quality Control

A large number of customers have said that the product they received was not the same as what was shown on the Daraz website or app. Sometimes the product is fake, sometimes it is damaged during handling, and sometimes it is simply of a much lower quality than what was expected from the pictures and description shown online. This usually happens because Daraz works with thousands of different sellers, and not every seller is checked carefully before they are allowed to sell on the platform. Because of this, some sellers list wrong pictures, misleading descriptions, or low-quality items just to attract buyers with a low price. In some reported cases, criminals have even created fake pages that look like official Daraz stores to sell counterfeit goods and collect advance payments without ever shipping a real product. This problem does not just affect the one customer who receives a bad product. It also damages the overall reputation of Daraz, because many buyers do not separate the platform from the individual seller in their mind. If they get a fake or low-quality item, they blame Daraz as a whole rather than just that one seller. Over time, repeated cases like this can push customers toward other shopping options, such as buying

directly from a shop, or through more informal channels like Facebook groups, even though those channels can carry their own risks.

Problem 02: Refund and Return Process

Many customers have complained that getting a refund from Daraz takes a very long time, or sometimes does not happen at all. In some cases, a customer returns a product, the delivery person collects it, but the customer never receives any paper or message confirming that the return was accepted. Later, the customer finds out that the return request was cancelled, without any clear reason given to them. Other customers report that even when a delivery clearly failed, or a product clearly arrived damaged, their refund request was still rejected, with no easy way to reach someone who could review the decision.

This creates a lot of confusion and frustration among customers, because they feel like their money is stuck with no clear or fair way to get it back. Since Cash on Delivery is common in Bangladesh, many customers have already paid real cash for a product before finding out it needs to be returned, which makes a slow refund process feel even more unfair. A slow and unclear refund process also adds extra pressure on the customer support team, since more people call in or message to ask about their pending refunds, which in turn makes support response times even slower for everyone.

Problem 03: Account Security and Fraud

Some customers have reported that their Daraz accounts were accessed without their permission, and this unauthorized access was then used to make purchases they never approved. In one such case, a customer's account was used to buy a Daraz gift card, which was then sent to an email address the customer did not recognize, and Daraz was not able to cancel the order or recover the gift card once it was claimed. Apart from direct account hacking, there have also been separate and well documented cases where criminals created fake Daraz websites and fake Facebook pages that looked almost exactly like the real Daraz brand, and used them to trick people into paying for products that were never sent. On a larger scale, a financial review carried out by a government body in Bangladesh found signs of unusual and suspicious money transactions connected to several e-commerce platforms operating in the country, including Daraz Bangladesh, over a multi-year period. Separately, industry reports have noted that online shopping fraud complaints in Bangladesh have been rising quickly in recent years, with scammers increasingly targeting accounts linked to mobile financial services such as bKash. All of these cases together, from individual account hacking to platform impersonation to wider financial fraud patterns, show that stronger account security, better fraud detection, and clearer steps to stop fake-website scams are needed to protect customers and preserve trust in the platform.

Problem 04: Delivery Service

Delivery delay is one of the most common complaints about Daraz. Many customers say their order took much longer to arrive than the time shown on the app at the moment they placed the order. In some cases, delivery riders mark an order as "delivery attempt failed" or say the customer was "not reachable", even though the customer says they were actually available and waiting for the product the whole time. This creates a mismatch between what the app says happened and what the customer says actually happened, which is hard for the customer to challenge or correct after the fact.

While Daraz has made real progress over the years, cutting average delivery times from around seven to nine days down to about two to three days in many areas by building its own logistics network and partnering with courier companies, delivery problems have not disappeared completely. Bangladesh's crowded city traffic, difficult weather during certain seasons, and harder-to-reach rural areas all continue to create delays from time to time. Because delivery is the last and most visible step of the whole shopping process, any mistake here has a very strong effect on how a customer feels about the entire company, even if every other step, such as browsing, ordering, and payment, went smoothly.

Problem 05: Customer Support and Seller Communication

Several customers have said that it is difficult to reach a real person at Daraz customer support when they have a problem. Many describe getting stuck with an automatic chatbot that gives the same repeated answers without actually solving the issue, forcing them to repeat their problem multiple times. In some cases, the support chat is closed by the company before the customer's problem is even solved, leaving the customer with no clear next step to take.

On top of this, communication with individual sellers is often slow, especially when a customer wants to ask about an order change, a missing item, or a delay in shipping. Since Daraz is a marketplace with thousands of independent sellers, the quality of communication can vary a lot from one seller to another, and Daraz itself has limited direct control over how quickly a seller responds. Poor communication at both the platform level and the seller level leaves customers feeling ignored and unheard, especially at the exact moment when something has already gone wrong with their order and they need help the most.

1.3 Conclusion

This chapter has looked closely at Daraz Bangladesh, starting with how the company works, its ownership under the Alibaba-owned Daraz Group, its mission and vision, its organizational structure, its payment and delivery systems, and what makes it such an important and widely used online shopping platform in Bangladesh. It has also pointed out five major problems that affect the day-to-day experience of Daraz customers: issues with fake or low-quality products, a slow and unclear refund process, weaknesses in account security and fraud protection, unreliable delivery service, and poor customer support and seller communication.

Each of these problems, on its own, might seem like a small issue. But because Daraz serves such a massive number of customers every single day, even a small percentage of failed orders, delayed refunds, or hacked accounts adds up to a very large number of unhappy customers in total. Left unaddressed, these problems can slowly weaken customer trust and loyalty, even for a company that holds a strong and largely unchallenged position in the Bangladeshi e-commerce market.

Finding and clearly writing down these problems is the first and most important step of system analysis. Without knowing exactly what is going wrong, it is not possible to design a better system. The problems identified in this chapter will be studied in more detail in the coming chapters. Chapter 2 will look at the initial feasibility of solving each of these problems, considering technical, economic, operational, schedule, and legal factors. Chapter 3 will go deeper into analysis using information gathering tools, charts, and diagrams, along with a costbenefit and break-even analysis. Chapter 4 will then present a redesigned system that aims to directly fix the problems described above, helping Daraz Bangladesh serve its customers in a safer, faster, and more trustworthy way.

Chapter 2
Initial Feasibility Study

2.1 Introduction

Recognizing a need is only the first step in system analysis. Once we know what the problems are, the next step is to check whether it actually makes sense to try and fix them. This step is called Initial Feasibility Study. It is an early check that helps us understand, in a simple way, whether a proposed solution to each problem is realistic, useful, and worth the time and money it would take.

In this stage, a system analyst usually works closely with different people connected to the system. This can include end users such as customers, managers, technical staff, and sometimes

even the sellers who use the platform. Talking to these people, along with reading reviews, reports, and other public information, helps the analyst understand the real requirements and expectations behind each problem. Both functional needs, meaning what the system should be able to do, and non-functional needs, meaning things like speed, safety, and reliability, are considered at this stage.

2.2 Initial Feasibility Analysis

Initial feasibility analysis examines whether the proposed solutions are realistic and practical from different perspectives, including technical and economic feasibility.

1.Technical Feasibility:

This checks whether the technology needed to fix the problem already exists, or whether it can be built or bought without too much difficulty. It looks at things like software, hardware, skilled workers, and any technical limits that might make the solution hard to build.

2.Economic Feasibility:

This is also called cost-benefit analysis. It compares how much money the solution will cost to build and maintain, against how much benefit or saving it will bring. If the benefits are clearly bigger than the costs over time, the solution is considered economically worth doing.

3.Operational Feasibility:

This checks whether the new solution will actually fit into the way the company already works. It looks at whether staff and users will be able to use the new process easily, and whether it matches the daily operations and goals of the business.

4.Legal Feasibility:

This checks whether the solution follows all the necessary laws and rules. For an e-commerce company like Daraz, this can include data protection laws, consumer protection rules, and payment or financial regulations set by the government or the central bank.

5.Schedule Feasibility:

This checks whether the solution can realistically be built and put in place within a reasonable amount of time. It looks at deadlines, how many resources are available, and any risks that could cause delays.

Using these five simple checks, we now look at each of the five problems identified in Chapter 1. For every problem, we state the problem again in short, summarize what we found, explain the findings in more detail, and finally give a clear recommendation.

The following sections analyze these feasibility aspects in relation to the major problems identified in the Daraz Bangladesh system

2.2.1 Problem 01 : Product Authenticity and Quality Control

2.2.1.1 Statement of the problem

Many customers who order products on Daraz Bangladesh receive items that are fake, damaged, or of much lower quality than what was shown on the website or app. This mainly happens because thousands of different sellers list products on Daraz, and not all of them are checked carefully before they are allowed to sell.

2.2.1.2 Summary of findings

The main issue is that Daraz does not have a strong enough system to check sellers and their products before allowing them to sell on the platform. Because of this, some sellers post wrong photos, wrong descriptions, or send products that are not what the customer expected. This lowers customer trust and can push buyers toward other shopping options.

2.2.1.3 Details of findings

Daraz Bangladesh works as a marketplace, which means the company itself does not make or directly control most of the products sold on the platform. Instead of thousands of independent sellers, ranging from small home businesses to large shops, list their own products and manage their own stock. This model helps Daraz offer a very large number of products, but also creates a problem . It becomes hard to check the quality and truth of every single listing.
Customer reviews collected from independent websites show a repeating pattern. Buyers often say that the product they received does not match the picture or description on the app. Some say the item looked used or damaged when it arrived, even though it was listed as brand new. Others report receiving a completely different item than what they ordered, or a cheap copy of a well-known brand instead of the real product. In a few serious cases reported in the news, criminals created fake pages that looked like real Daraz seller stores, took payment from customers, and never sent any product at all.

This kind of problem is common in large scale online marketplace around the world, not just in Bangladesh, because it is genuinely hard to check millions of products from thousands of sellers one by one. However, the impact on trust is still very real. When a customer receives a bad product, they often do not blame the seller. Instead, they blame Daraz as a company, since Daraz is the platform they used and trusted to place the order. This means that even one bad seller can create a bad experience that affects how a customer feels about the whole Daraz brand.

Another part of the problem is that even when Daraz has some rules for sellers, such as warnings or account suspension for repeated complaints, this information is not always clear to customers

before they buy. A customer usually only finds out that a seller has a bad reputation after they already receive a bad product. If there was a stronger and more visible system to check and rate sellers before a purchase, many of these bad experiences could be avoided in the first place.

Looking at this problem through the five feasibility checks explained earlier, building a stronger seller-verification and product-quality system is technically possible, since many similar marketplaces around the world already use tools like ID verification, seller scorecards, and random quality audits. It is also operationally realistic, because it builds on top of the seller-management system Daraz already has, rather than replacing it completely. The main cost would come from hiring more staff or building automated tools to review sellers and products, but this cost is likely to be smaller than the long-term cost of losing customer trust. From a legal point of view, stronger checks on sellers would actually help Daraz follow consumer-protection expectations more closely, which lowers legal risk rather than raising it. Overall, this problem looks feasible to address within a reasonable schedule, especially if improvements are rolled out in stages, starting with the most complained-about product categories.

2.2.1.4 Recommendation

- Build a stronger seller verification process before a new seller is allowed to open a shop on Daraz, checking basic identity and business information.
- Use a clear seller rating and badge system, so customers can easily see which sellers have a strong history of good reviews before they buy.
- Add random quality checks on best-selling and newly listed products, especially in categories like electronics and beauty products where fake items are more common.
- Make it easy for customers to report fake or wrong products directly from the order page, and act quickly on sellers who receive many such reports.
- Clearly warn customers when a product or seller has a history of complaints, instead of only showing this information after a problem happens.

2.2.2 Problem 2 : Refund and Return Process

2.2.2.1 Statement of the problem

Daraz customers often face long delays or unclear answers when they try to get a refund or return a product. In some cases, a return is collected by the delivery rider, but the customer never gets a clear confirmation, and the request is later cancelled without a proper reason.

2.2.2.2 Summary of findings

The refund and return process at Daraz lacks clear steps and fast communication. Customers are often left waiting without knowing the real status of their money or their returned item. This creates frustration and a feeling that the process is unfair, especially for customers who paid cash on delivery .

2.2.2.3 Details of findings

A return and refund process usually has a few clear steps: the customer requests a return, the company approves or rejects it, the product is picked up, the product is checked, and finally the money is refunded. In the case of Daraz, customer reviews show that this chain of steps often breaks somewhere in the middle. Some customers report that the item was picked up by a rider, but they were never given a signed paper, a receipt, or even a simple message confirming that the return was received by the company.

Later, some of these same customers say they checked their account and found that the return request had been marked as "cancelled", with no clear explanation of why. This is a serious gap, because from the customer's side, they gave away their product in good faith, but have no strong proof that this happened, and no clear next step to take if something goes wrong. Other customers describe cases where a delivery clearly failed, or a product clearly broken, but their refund request was still rejected without a proper review.

Because Cash on Delivery is very common in Bangladesh, many customers have already handed over real cash before they realize a product needs to be returned. This makes a slow or unclear refund process feel even more unfair and stressful, since it is not just a digital number changing on a screen, but real money the customer is waiting to get back. When customers do not trust that a refund will actually happen, they may become hesitant to order certain types of products again, especially higher-priced items.

The unclear refund process also creates extra pressure inside the company itself. When customers do not get clear updates, many of them contact customer support again and again to ask about their refund status. This adds more work to a support team that may already be dealing with many other requests, which can slow down response times for everyone, not just the customers with refund problems.

From a feasibility point of view, fixing the return and refund process is one of the more straightforward problems to solve. Technically, adding status tracking and automatic messages to the return process uses the same kind of technology Daraz already uses for order and delivery tracking, so no completely new system needs to be built. Economically, the cost of adding these features is likely to be low compared to the savings gained from fewer repeated customer-support calls about refund status. Operationally, clearer return steps would actually make life easier for support staff too, since they would spend less time answering the same "where is my refund" questions again and again. There are no major legal barriers to making a refund process clearer and faster; if anything, clearer timelines help Daraz meet consumer-protection expectations. Overall, this problem can realistically be improved within a short to medium timeframe.

2.2.2.4 Recommendation

- Give customers a clear, trackable status for every return, similar to how an order delivery is tracked, so they always know exactly what stage their return is at.

- Send an automatic message or digital receipt the moment a rider picks up a returned item, so the customer has proof the return was received.
- Set a clear maximum time limit for refunds to be completed, and share this time limit with customers so they know what to expect.
- Explain the exact reason whenever a return or refund request is rejected or cancelled, instead of just changing the status with no message.
- Create a simple way for customers to ask for a second review if they feel their refund was unfairly rejected.

2.2.3 Problem 03: Account Security and Fraud

2.2.3.1 Statement of the problem

Some Daraz customers have had their accounts accessed without their permission,leading to purchases they never approved.There are also separate cases of fake websites and fake social media pages pretending to be Daraz, which are used to steal money from unsuspecting buyers.

2.2.3.2 Summary of findings

Daraz faces two connected but different security risks: direct account hacking,where a real customer account to broken into and impersonation fraud ,where criminals create fake pages that only look like Daraz.Both problems reduce customer trust and put people's money at risk and both point to a need for stronger digital safety measures.

2.2.3.3 Details of findings

Account security is one of the most serious issues for any platform that handles both personal information and payments . In one reported case, a Daraz customer's account was accessed by someone else without permission and this access was used to buy a Daraz gift card.This gift card was then sent to an email address the account owner did not recognize and by the time the issue was noticed, the company was not able to cancel the order or take back the gift card .This shows that once an account is broken into,the current system does not have a fast enough way to stop or reverse the damage . Separately from direct hacking, there is also a pattern of fraud where criminals build fake websites with names very close to the real Daraz website or create fake Facebook pages using the Daraz name and logo. These fake pages are used to trick customers into paying for products that do not exist or that will never actually be shipped. Because these fake pages can look very convincing ,many ordinary customers cannot easily tell the difference between the real Daraz and a fake copy,especially when the offer looks like a good discount.

On a bigger scale, a review carried out by a Bangladeshi financial authority looked into suspicious money transactions connected to several e-commerce platforms operating in the country,including Daraz, over a period of a few years .This shows that account and payment

security issues are not only a small, personal problem for a few unlucky customers, but also something that has drawn attention at a national ,regulatory level. Separately ,general reports on online fraud in Bangladesh show that scam complaints related to online shopping have been rising quickly with criminals increasingly targeting accounts that are connected to mobile financial services such as bKash.

All of these different pieces, from one hacked account, to fake websites, to wider financial-fraud patterns, point to the same root cause : the current system does not have strong enough tools to detect unusual activity early, confirm that a person trying to log in or make a purchase is really who they claim to be or clearly teach customers how to tell a real Daraz page from a fake one. Checking this problem against the 5 feasibility areas,account security improvements are technically well understood , since tools such as one time SMS codes and automatic fraudflagging are already common across banking and mobile-financial service apps in Bangladesh ,so Daraz would not be starting from 0 .The economic cost of adding these tools in real investment,but it is small compared to the potential cost of losing customer trust on a national scale ,especially after a regulatory body has already raised concerns.Operationally ,extra security steps need to be added carefully so they do not make the checkout process too slow or annoying ,but this is a manageable design challenge .Legally ,stronger fraud protection actually helps Daraz stay aligned with financial sector expectations in Bangladesh ,rather than creating new legal risk .Given the seriousness of this problem , it deserves to be treated as a high priority within the project schedule, even if it takes a bit more time and coordination with the outside partners such as banks and mobile-financial-service providers .

2.2.3.4 Recommendations

- Add extra login security , such as sending a one-time code by SMS whenever someone logs in from a new device or tries to change account details .
- Build a fraud detection system that flags unusual activity , such as a sudden high-value purchase or a login from an unexpected location,before the order is completed.
- Given customers a fast and simple way to freeze their account and stop pending orders the moment they suspect their account has been hacked.
- Run regular public awareness campaigns to teach customers how to recognize the real Daraz website and app and how to avoid fake pages and links.
- Work closely with banks ,mobile financial service providers and government bodies to quickly report and act on suspicious transaction patterns.

2.2.4 Problem 04: Delivery Service

2.2.4.1 Statement of the problem

Daraz customers often experience delivery delays that go beyond the time shown at checkout. In some cases,delivery riders mark an order as failed or the customer as unreachable , even though the customer says they were available and waiting the whole time.

2.2.4.2 Summary of findings

While Daraz has improved its overall delivery speed a lot over the years by building its own logistics network and working with courier partners , delays and wrong status updates still happen. This creates a gap between what the app tells the customer and what is actually happening with their order ,which is one of the most visible and frustrating types of problems for any online shopper.

2.2.4.3 Details of findings

Delivery is often described as the final and most important step of the online shopping experience,because it is the moment a customer actually receives what they paid for. In the early years of e-commerce in Bangladesh, delivery times ere often quite long,sometimes taking around 7 to 9 days for a single order to arrive .Over time, Daraz built its own delivery network and also partnered with third-party courier companies to speed things up.Beacuse of these changes ,average delivery time in many areas has dropped to around 2 to 3 days,which is a big improvement .

However ,even with this progress ,delivery problems have not fully gone away. Customer reviews describe cases where an order simply takes much longer than the estimated delivery rate shown at the time of purchase ,without any clear updated explanation from the company .Other reviews describe a more specific and frustrating problem: the delivery status shows that a rider attempted delivery and could not reach the customer or that the customer was " unreachable " , while the customer insists they were home and available the entire time, sometimes even keeping their phone nearby all day waiting for the delivery call.

This mismatch is difficult for a customer to challenge after the fact,since there often no simply way to prove what actually happened during a failed delivery attempt .From the customer's side ,this can feel like they are being blamed for a problem that was not their fault,which adds to their frustration on top of simply not receiving their order on time .Since a large country like Bangladesh has crowded city traffic,seasonal weather problems and harder to reach rural areas ,some delays are naturally expected but customers still expect clear and honest communication when something goes wrong , rather than a status update that does not match reality.

Because delivery is the last thing a customer expectations in their shopping journey , a bad delivery experience can undo a lot the good will built up during browsing,ordering payments,even if those earlier steps went smoothly.This makes delivery reliability one of the

most important areas for Daraz to keep improving ,even though real progress has already been made compared to the early days of the company.

Looking at delivery through the 5 feasibility checks, this problem is technically feasibility to improve ,since Daraz already has a working logistics network and simply needs better proof-of-attempts tools and clearer rider accountability ,rather than a completely new delivery system . Economically ,the cost of adding simple tools like photo proof or call logs is small,especially compared to the cost of losing repeat customers due to bad delivery experiences. Operationally, this change mainly affects how riders record their work,which means it can be added through training and a small update to the rider app,without disrupting the whole delivery network .There are no major legal issues involved in improving delivery accuracy. Given that Daraz has already shown it can make big improvements to delivery speed in the past ,further improvements to delivery accuracy and communication are realistic within a reasonable schedule.

2.2.4.4 Recommendation

- Give delivery riders a simple way to log proof of a failed delivery attempt , such as a short note,a call log or a photo,instead of only a status label.
- Allow customers to reschedule a delivery time themselves through the app if they know they will not be available at a certain time.
- Send real- time delivery updates with a working contact number for the rider,so customers and riders can coordinate directly if there is any confusion.
- Track and review delivery riders who have a high number of "failed delivery" or "Unreachable customer" reports , to check if this is a pattern rather than a one-time mistake.
- Continue to expand the logistics network and courier partnerships in harder-to-reach areas outside of the major cities.

2.2.5 Problem 05:Customer Support and Seller Communication

2.2.5.1 Statement of the problem

Many Daraz customers find it hard to reach a real support agent when they have a problem ,often getting stuck with an automatic chatbot that does not solve their issue.Communication with individual sellers is also often slow,especially when a customer needs help with an order problem.

2.2.5.2 Summary of findings

Customer support at araz relies heavily on automatic chat systems, which can feel un helpful when a customer has a specific or unusual problem. At the same time, since Daraz works with

many independent sellers,the speed and quality of communication can vary a lot from one seller to another, and this is harder for Daraz to fully control.

2.2.5.3 Details of findings

Good customer support is one of the most important parts of any online shopping platform,because it is the main way a company can fix a problem after something has already gone wrong.Beased on customer reviews, a common complaint about Daraz is that reaching a real human support agent is difficult. Many customers describe being met first by an automatic chatbot, which asks simple preset questions and gives per-written answers. For simple issues,this can work fine, but for more specific or unusual problems,customers often feel like they are going in circles without getting real help.

Some customers report an even more frustrating pattern, where a support chat is closed by the company side before their actual problem has been solved.This leaves the customer needing to start the whole conversation over again, sometimes repeating the same details multiple times to different support staff or bots. This kind of repeated effort can make a customer feel unheard, especially if they are already upset about a separate problem, such as a late delivery or a damaged product.

On top of platform-level support, communication with individual sellers is a separate but related issue.Since Daraz is a marketplace with thousands of independent sellers, each seller manages their own communication with buyers,usually through in app chat. Some sellers reply quickly and helpfully, while others are slow to respond or do not reply at all, especially once a sale has already been completed. This creates an inconsistent experience, where a customer's satisfaction can depend heavily on which specific seller they happened to buy from, rather than on Daraz as a whole company.

Because Daraz has limited direct control over how quickly a seller replies, this part of the problem is harder to fix compared to Daraz's own support team. However, Daraz can still play a stronger role in tracking seller response times and setting clear expectations for how fast sellers should respond to buyer messages. Without this kind of oversight,poor seller communication continues to reflect badly on the whole Daraz platform, even when the fault lies mainly with one individual seller.

From a feasibility standpoint, improving customer support is realistic and does not require any unusual or experimental technology.Technically, adding an easier path to a human agent and tracking seller response times can be built on top of the chat and messaging systems Daraz already runs. Economically,better support may need a bit more staffing or smarter routing tools, but this cost is balanced out by fewer repeated support tickets and happier,more loyal customers over time.Operationally, clearer rules about when a chat can be closed, and expected seller response times are simple policy changes rather than large technical projects, which makes them

easier to put into practice quickly.There are no legal barriers to improving support quality.Overall, this is one of the more achievable problem to fix, and improvements here could start showing results in a relatively short amount of time.

2.2.5.4 Recommendation

- Give customers a clear and easy way to reach a real human support agent, especially after a chatbot has failed to solve their problem twice.
- Set a rule that a support chat cannot be closed by the company side unless the customer confirms their issue is actually solved.
- Track and publicly display average seller response times, so customers can see this information before they buy from a seller.
- Set a minimum expected response time for sellers, and follow up with sellers who repeatedly reply too slowly to buyer messages.
- Offer multiple ways to contact support, such as chat, phone, and email, so customers can choose the option that works best for their specific problem.

2.2.6 Overall Feasibility Summary

Before moving to the conclusion, it is helpful to put all five problems side by side and see how they compare across the five feasibility areas discussed earlier in this chapter. The table below gives a simple, high-level view of this comparison, based on the details of findings explained above.

Problem	Technical	Economic	Operational	Legal	Schedule
Product Authenticity and Quality Control	Feasible	Moderate cost	Feasible	Low risk	Medium term
Refund and Return process	Highly feasible	Low cost	Highly feasible	Low risk	Short term
Account Security and Fraud	Feasible	Higher cost	Feasible	Low risk	Medium term
Delivery service	Highly Feasible	Low cost	Feasible	Low risk	Short term
Customer support and seller Communication	Highly Feasible	Low cost	Highly Feasible	Low risk	Short term

Table 2.1: Simple feasibility comparison across the five identified problems

As the table shows, none of the five problems were found to be unrealistic or impossible to fix. The refund and return process, delivery service, and customer support problems all look easier and faster to improve, mainly because they build on systems Daraz already has in place. Product authenticity and account security are a little more demanding, since they need extra tools such as fraud detection and stronger seller checks, but they are still realistic to achieve with proper planning and a reasonable budget.

2.3 Conclusion

This chapter has looked at five main problems identified in chapter 1 through the lens of initial feasibility analysis. For each problem , we explained the situation in more detail, described the patterns found in customer reviews and public reports, checked the problem against technical, economic, operational, legal, and schedule feasibility, and gave clear, practical recommendations that could help improve the situation.
Across all five problems, a few common themes appear again and again. Many of the issues come from the fact that Daraz operates at a very large scale, connecting millions of customers with thousands of independent sellers and a nationwide delivery network. At this scale, even a small page in checking communication, or security can affect a very large number of people. At the same time, the recommendations given in this chapter show that most of these problems can be improved without needing a completely new business model. Many of the fixes are about adding clearer checks, faster communication, and better use of technology, such as automatic status updates, stronger fraud detection, and clearer seller ratings.

It is also worth noting that these five problems are not fully separate from each other. For example, weak seller verification (Problem 1) can make fake-product complaints harder to resolve through the refund process (Problem 2). Similarly, weak account security (Problem 3) can lead to disputed orders that then create extra pressure on customer support (Problem 5). This means that solving one problem well can often make it easier to solve another, which is a useful thing to keep in mind when planning which improvements to build first.

Based on this initial feasibility check, all five problems appear realistic and worth pursuing further. None of them require something impossible or extremely rare in the e-commerce industry; in fact, many similar platforms around the world already use some version of the solutions recommended here. This means the project can move forward with confidence into the next stage. Chapter 3 will

now go into a deeper analysis of the existing system, using information gathering tools, charts, a cost-benefit and break-even analysis, and a closer look at the root causes behind these problems.

Chapter 3
Analysis

3.1 Introduction

Information gathering and detailed analysis are some of the most important steps in system analysis.This stage is about collecting,organizing and studying information about the system so that we can fully understand how it works,what its requirements are and what problems exist inside it.Without good information, it is not possible to make good decisions about how to improve a system.

The main goal of this stage is to find the real problems and opportunities that need attention.This usually means looking closely at how the current system works,how users interact with it and what data or reports are already available about it.A good analyst tries to collect information from many different angles so that the final picture is complete and fair,instead of relying on just one point of view.

This chapter follows that same approach for Daraz Bangladesh.Instead of relying on a single source of information,several different tools are combined so that the strengths of one tool can help cover the gaps left by another.A review of published literature and procedures gives broad,industry-level context;direct observation of the platform show how the system behaves moment to moment; a real call to Daraz's own customer care helpline adds a first-hand, human data point; and a structured analysis of public customer reviews adds scale, by turning many individual experiences into patterns that can be compared and charted.

By carefully gathering and studying information,we are able to build a clear and realistic picture of how Daraz Bangladesh's ordering,payment,delivery and support system works today.This understanding is the base on which the rest of this chapter and the design work in Chapter 4,will be built.

3.2 Information Gathering

Gathering information is a key part of the analysis process. It requires both a scientific side and a creative side.The scientific side means using organized methods and tools so that information is collected in a fair and structured way. The creative side means using good judgment to

understand what the information actually means and to notice patterns that are not always obvious at first glance.

There is no single fixed way to gather information for every project.However, one rule the same in every case;information should be collected carefully,honestly and in a way that causes as little disturbance as possible to the people or systems being studied.For this report,four main tools were used to gather information about Daraz Bangladesh's system.These four tools are Review of literature,procedure and forms;On-site observation;Interview; and Questionnaires.Each of these tools is explained in sections that follow.

It is worth being upfront about one practical limit: reaching Daraz Bangladesh's senior management directly for a formal interview is not realistic for a student project.Because of this, the interviews section in this chapter is built around a channel that is genuinely reachable,Daraz's own customer care helpline, combined with information already published by the company itself.This keeps the information gathering process honest and grounded in real,verified channels rather than assuming access that a student group would not normally have.

3.2.1 Review Of Literature,Procedure And Forms

Reviewing literature,procedure and forms is one of the most useful early steps in information gathering . It means looking at existing knowledge, common industry practices and the standard documents or screens already used inside a system. This step builds a strong foundation before moving on to more direct methods such as interviews or questionnaires.

Review Of Literature

The review of literature means studying general knowledge,industry standards and research related to the field being analyzed.For Daraz Bangladesh ,this means looking closely at trends in e-commerce marketplace,both in Bangladesh and around the world,to understand what " good practice" looks like in this industry.

Industry Standards

Large online marketplaces around the world,such as Amazon,Alibaba and Shopee, generally follow a set of common standards to keep customers safe and happy.These standards include verifying sellers before allowing them to sell,offering buyer protection programs that guarantee a refund if a product does not arrive or does not match its description and using automated systems to detect fraud before an order is completed.Studying these standards helps us understand where Daraz Bangladesh's current practices might be falling behind what is now considered normal in the wider e-commerce industry.

Best practices

Looking at how other successful e-commerce platforms manage their operations gives useful ideas for improvement.For example,many platforms use a clear “seller score” system,where a seller’s rating,response time and order-cancellation rate are shown directly to customers before they buy.Some platforms also use real-time chat support with a guarantee maximum wait time and automatic refunds for very common and simple problems and its customers and are useful models to compare against Daraz Bangladesh’s current system.

Technology Advancements

New technology is charging how e-commerce platforms operate. Many platforms now use artificial intelligence to predict which products are likely to be fake based on price , seller history and product and product category. Real-time GPS tracking is now common for delivery, allowing customers to see exactly where their rider is on a map. Automated fraud-detection systems can flag unusual account activity ,such as a sudden login from a new device followed by a large purchase, within seconds.These kinds of technology can offer useful direction for how Daraz Bangladesh could strengthen its own system.

Comparison With Other Platforms Operating In Bangladesh

Looking briefly at how other platforms operating in Bangladesh,such as Pathao Food,Foodpanda and Chaldal,handle similar problems is useful context for this review.Ride-share and delivery platforms like Pathao generally give customers a way to directly message or call their assigned rider,something Daraz’s delivery tracking does not currently offer.Grocery platforms like Chaldal,which manage their own stock rather than relying on third-party sellers are able to check product quality more directly before an order ships,which is harder for Daraz to do given its much larger,marketplace-based catalog.This kind of comparison shows that some of Daraz’s problems come from the specific tradeoffs of running a large,open marketplace,rather than a lack of technology alone;even so,several of the practices used by these other platforms could still be adapted to fit Daraz’s own system.

Review Of Procedures

Reviewing internal procedures means looking closely at the steps a company its daily work.Since it is not possible to sit inside.Daraz Bangladesh’s own operations meetings,this review is based on how the process appears from the outside,through the app,the website,public help-center pages and description the company has shared about its own operations.

Order Processing

The order process on Daraz begins when a customer searches for a product,adds it to their cart and proceeds to checkout.The customer then selects a payment method,which can be Cash on delivery ,a mobile financial service such as a bKash or Nagad or a card payment.Once the order

is placed,it is sent to the for confirmation and then prepared for handover to a delivery partner .Reviewing this process closely shows that most delays and confusion tend to happen either right after an order is placed, while waiting for seller confirmation, or during the final handover to the delivery rider.

Seller Onboarding

Sellers who want to open a shop in Daraz go through a registration process where they provide basic business details and list their products. Based on public information ,this process appears to focus mainly on getting sellers active quickly ,rather than doing a very deep check of the seller's history or the authenticity of their product.This matches what was found earlier in Chapter 1 and Chapter 2,where weak seller verification was identified as a root cause behind fake or low-quality product complaints.

Delivery Operations

Daraz coordinates delivery through its own logistics network and through courier partners such as Paperfly,REDX and eCourier. Order are grouped,handed over to riders and tracked through a delivery -status system inside the app.This process has clearly improved a lot over the years,cutting delivery times from around seven to nine days down to two to three days in many areas.However, the review of this procedure also shows that once a parcel handed over to a courier partner, Daraz's own visibility and control over exactly what happens during the last mile of delivery becomes more limited.

Customer Service Procedure

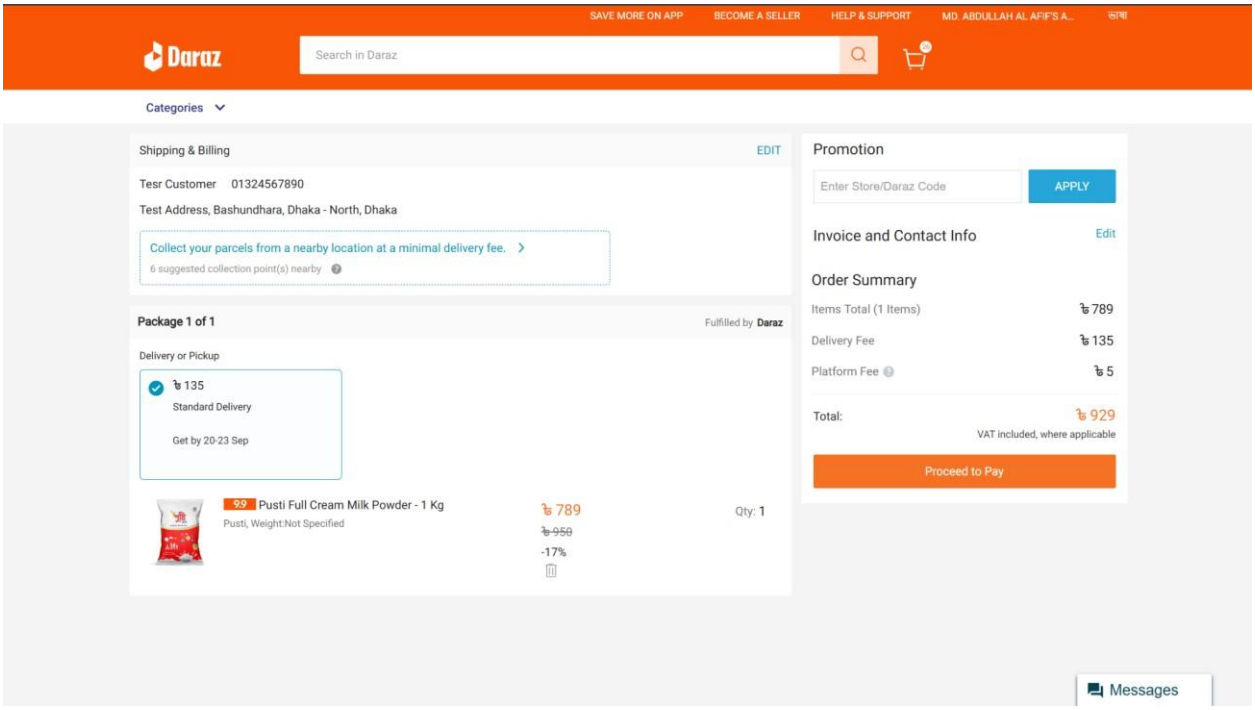
Customer service on Daraz is mainly handed through an in-app chat system,which usually starts with an automated chatbot before, in some cases,moving to a human agent.Daraz also runs a Central Complaint Management System (CCMS),a formal channel through which customers and sellers can raise complaints that need closer review .Reviewing this procedure shows that the process is built to handle a very large number of simple, repeated questions quickly,but it appears less well designed for handling more complex or unusual complaints that need a real person's judgment.

Review Of Forms

Forms,whether they appear as a digital screen or a printed document are an important part of how a company collects and manages data.Reviewing the forms used by Daraz Bangladesh helps us understand how information flows into and out of the system.

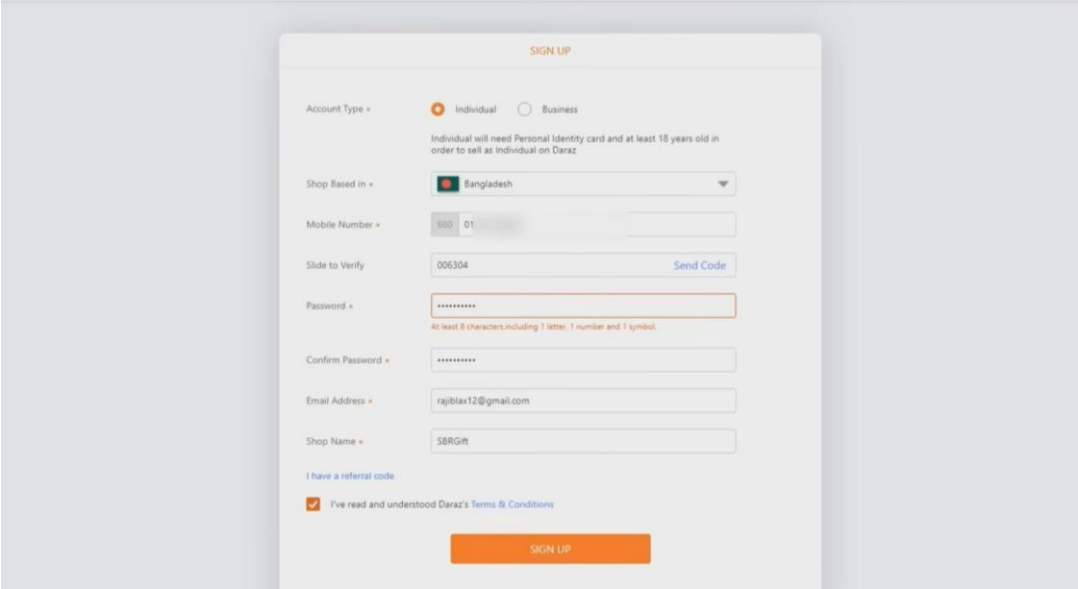
Order Form

The order form on Daraz collects the customer's delivery address,preferred delivery time slot,payment method and any applicable discount code.This form is generally easy to use but reviewing customer feedback shows that some customers want clearer confirmation right after submitting the form,since they are sometimes unsure whether their order actually went through ,especially during high-traffic sale products.



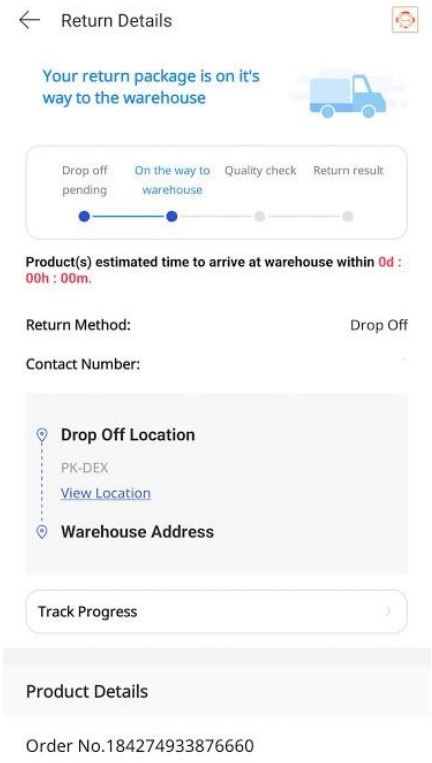
Seller Registration Form

The seller registration form collects basic business and contact information from a new seller before they can start listing products.Based on public information,this form does not appear to require heavy documentation or a long waiting period,which makes it easy for legitimate small business to join quickly,but may also make it easier for a small number of dishonest sellers to join without being properly checked.



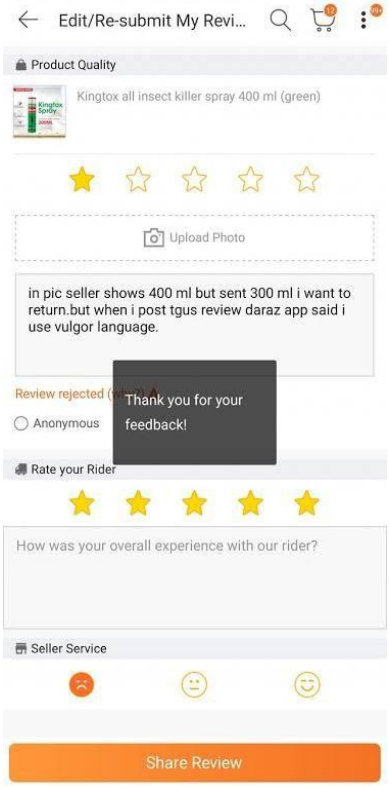
Return Feedback and Review

The return and refund form lets a customer choose a reason for their return,such as a wrong item,a damaged item,or a change of mind and then submit a request.Reviewing this form shows that while it is simple to fill out,it does not always give the customer a clear,trackable confirmation once submitted , which matches the refund and return problems already identified in Chapter 1 and Chapter 2 .



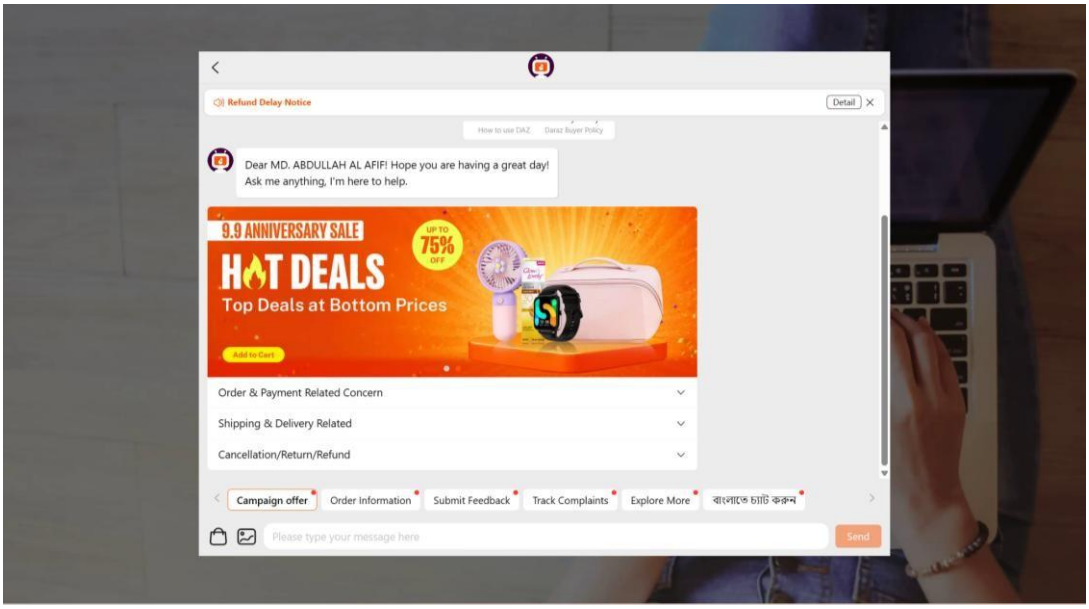
Customer Feedback and Review Form

After a delivery is completed,customers are able to leave a star rating and a written review for both the product and the seller.This form is useful for collecting customer opinions and it is also one of the main sources of information used this report ,since these public reviews are what allowed us to study patterns in customer satisfaction alongside the direct helpline contact described later in this chapter.



CCMS

Daraz Bangladesh also provides a separate ,more formal complaint form through its CCMS.This system is meant for issues that were not resolved through the normal chat based support process and it appears to be aimed at giving customers and sellers a more structured , trackable way to escalate a serious or unresolved problem.



Review Of Public Reports and Disclosures

Beyond the company's own material,useful information also comes public reports written by outside parties.Local business news outlets,such as The Business Standard,The Daily Star and Future Startup,regularly publish coverage of Daraz's market position,leadership changes and any notable incidents,including fraud cases involving fake websites that impersonate the Daraz brand.A financial review carried out by a Bangladeshi financial authority also looked into suspicious transaction patterns connected to several e-commerce platforms operating in the country ,including Daraz ,which is useful independent context for the account-security problem identified in Chapter 1.

Reviewing this kind of outside reporting ,alongside Daraz's own literature,procedures and forms , helps balance the picture.Company-published materials naturally focus on what the platform does well and how customers are meant to use it,while independent news coverage and regulatory findings are more likely to surface the problems that a purely internal or promotional source might not mention.

Taken together ,this review of literature ,procedures and forms gives a strong starting picture of how the Daraz Bangladesh system works today and it lines up closely with the problems identified in the earlier chapters of this report .This next series will build on this picture using onsite observation ,interviews and questionnaires.

3.2.2 On-Site Observation

On site observation is normally done by physically visiting a company's office or warehouse. Since a direct visit to Daraz Bangladesh's offices or warehouses was not possible for this

project,the observation for this report instead focuses on the platform itself,meaning the Daraz mobile app and website,which are the parts of the system that customers actually interact with everyday. Careful,hands-on use of the app and website,combined with attention to how orders, payments, and deliveries are shown to the user, gives useful insight into how the system behaves in real use.

The goal of this observation was to check how smoothly a customer can move through the full shopping journey, from opening the app to receiving a product ,and to notice any points where the system seems unclear, slow, or inconsistent. The following points summarize what was observed.

1.Home Page and Product Browsing

The Daraz home page is built around large banners for ongoing sales, category shortcuts, and personalized product suggestions. Searching for a product is fast, and filters for price, brand, and rating are available. However, product listing from different sellers for the same item can appear side by side with very different prices and photo quality, which can make it hard for a customer to judge which listing is trustworthy just by looking at the same results

2. Product Page and Seller Information

Each product page shows the price,a description, customer reviews, and a rating for the seller. The seller's overall rating is visible, but detailed information such as how long the seller has been active, or how many of their orders have led to return or complaint,is not easy to find. This matches the seller-verification concerns raised in chapter 1 and 2.

3. Checkout and Payment Flow

The checkout process is short and clearly laid out, asking for a delivery address, a delivery time preference, and a payment method. Cash on Delivery is offered as a simple default option, alongside mobile financial services and card payment.The final order summary clearly shows the total price, but does not always clearly explain the store's specific return policy for that particular product before the order is confirmed.

4. Order Tracking and Delivery Status

Once an order is placed,the app shows a step-by-step status, such as "Order Confirmed" , "Packed" "Shipped", and "Out for Delivery". This tracking system is generally clear and easy to follow. . However, in line with the delivery complaints discussed in earlier chapters,there is no built-in way for a customer to directly message their assigned rider, or to see proof of a failed delivery attempt if the status suddenly changes to "delivery attempt failed".

5. Return, Refund, and Customer Support Flow

Starting a return inside the app is simple: a customer selects the order, chooses a reason,and submits the request. After that point, though, the app does not always show a very detailed, stepby-step status for the return, unlike the clear tracking given for a normal delivery. Reaching customer support is done through an in-app chat, which usually starts with an automatic chatbot offering a set of common topics before a customer tries to reach a human agent, or escalates further through the Central Complaint Management System mentioned in section 3.2.1.

6. Notifications and Ongoing Communication

Daraz sends push notifications and SMS messages for major order events, such as order confirmation, shipment, and delivery. These notifications are generally timely for the early stages of an order.However, once an issue comes up, such as a delayed shipment or a return request, the frequency and clarity of follow-up notifications becomes less consistent, leaving the customer to check the app manually rather than being kept updated automatically.

Overall, this observation shows that the front-facing, customer-side parts of the Daraz system are generally well designed and easy to use for a normal, problem-free order. The weaker points appear mainly in the areas already flagged in Chapters 1 and 2: seller trust information,delivery-status accuracy, and the depth of tracking and communication available once something goes wrong with an order.

A short summary of what each information-gathering tool contributed to this report is given in the table below.

Tool	Main Contribution to This Report
Review of Literature, Procedure, and Forms	Gave a general picture of how Daraz's ordering, delivery, and support systems are structured, and how they compare with wider e-commerce industry practices.
On-Site Observation (Platform)	Showed exactly how the app and website behave in real use, and pinpointed specific screens and steps where the system is weaker.
Interviews (Helpline Call and Public Materials)	Added a direct, real-world channel of information through Daraz's own customer care line, plus context from the company's published policies
Questionnaires and Review Analysis	Turned a large number of individual public reviews into structured, comparable data, shown as the charts in Section 3.3.

Table 3.1: Summary of what each information-gathering tool contributed to this chapter.

3.2.3 Interviews

For this report, the interview process was built around a channel that a student group can genuinely and honestly access: Daraz Bangladesh's own customer care helpline. Reaching senior company management directly was not realistic for this project, so instead of assuming that kind of access, the group placed a real call to Daraz's public customer care line.This section first gives the call details and a guide of the questions asked, with space for the actual notes taken during the call. It then adds supporting information drawn from Daraz's own published customer-service materials, which helps fill in details that a single support call may not fully cover.

Before placing the call, the group briefly considered which interview channel was realistic to use. The comparison below explains why the customer care helpline was chosen as the main directcontact method for this report.

Possible Channel	Possible Channel	Reason
Senior management / executives	Not realistic	Not normally accessible without prior corporate contacts or a formal media request
In-person office / warehouse	Not realistic	Requires prior company approval
visit		and is outside the group's access
Customer care helpline (16492)	Realistic	Open to the public, reachable by phone, designed to answer exactly these kinds of questions
Public reviews and published materials	Realistic	Freely available online and reflects real customer and company statements

Table 3.2: Comparison of possible interview channels and why the helpline was chosen.

3.2.3.1 Helpline Interview (Customer Care agent – Ritu Akter)

Daraz Bangladesh's customer care can be reached through the following official channels, as published on the company's own help-center pages:

- Hotline: 16492 (short code) or +8809610096111
- Hotline hours: Sunday to Thursday, 9:30 AM to 11:00 PM
- Email: customer.care@daraz.com.bd
- In-app Live Chat, available from the Help Center inside the Daraz app

#	Question Asked to the Support Agent	Findings

1	What should a customer do if a delivered product turns out to be fake, damaged, or different from what was ordered?	There are several options available for reporting regarding these issues. Customer can report from the app or website, also they can report by calling to 16492
2	How long does a refund normally take once a return is approved, and how is the customer informed about its progress?	Once a Daraz return is approved and processed, the refund usually takes 1–10 working days, depending on the payment method. Daraz generally updates customers through SMS/email, and you can also track the status in the Daraz app/website.
3	What should a customer do if they believe their account was accessed without their permission?	If you believe your Daraz account was accessed without your permission, please change your password immediately and secure the email/phone number linked to your account. Also check for any unauthorized orders or account changes.
4	If a delivery is marked as a failed attempt but the customer says they were available, what is the process to dispute or correct this?	Please share your order number with us. We'll raise a complaint and have the delivery attempt reviewed. If the order is still eligible for delivery, we'll assist with the next delivery attempt.
5	If a chat with customer support does not resolve an issue, what other steps or options are	If Live Chat doesn't resolve the issue, the customer can request escalation, follow up
	available to the customer?	using their case number, or contact Daraz Customer Care at 16492 for further assistance.
6	Roughly how many customer queries does the support team handle in a typical day, especially during sale campaigns?	There isn't a fixed daily figure; normal days have a steady volume, while major sale campaigns can bring a significant spike in customer queries.

Table 3.3: Helpline interview guide used to structure the call to Daraz Bangladesh customer care (16492)

Call log: Date of call — 26/08/2026. Time of call — 01:23 PM. Channel used — Hotline.

3.2.3.2 Supporting Information from Daraz's Published CustomerService Materials

Alongside the direct helpline contact, Daraz Bangladesh's own public help-center pages describe a number of standard policies that are useful context for this report. These are not answers from a live conversation, but they reflect what the company itself has published for customers, and they are presented here in a simple question-and-answer style for easy reading.

Q: What payment methods are available on Daraz?

A: Customers can pay using Cash on Delivery, mobile financial services such as bKash and Nagad or by card and online banking, depending on what is available for a specific order. **Q: How long does return usually take?**

A: Most products can be returned within a set number of days if they arrive damaged, incorrect, or significantly different from their description, though the exact return window can vary by product category and seller.

Q: What is the Central Complaint Management System (CCMS)?

A: It is a separate, more formal complaint channel for issues that were not resolved through the normal chat-based support process, meant to give customers and sellers a structured way to escalate an unresolved problem.

Q: Can a customer track the exact location of their delivery rider in real time?

A: Published help-center information describes step-based delivery status updates, such as "Shipped" and "Out for Delivery", rather than a live map-based rider location, which is a feature some competing delivery services in Bangladesh already offer.

Q: What happens if a product is out of stock after an order is placed?

A: In such cases, the standard process described by Daraz involves cancelling the affected item and notifying the customer, with any payment already made for that item returned through the normal refund process.

3.2.3.3 Seller Interview - Daraz Seller (Arman Rahman)

- **Interviewee:** Daraz Seller – Arman Rahman
- **Platform:** Daraz Bangladesh
- **Purpose:** To understand the seller's experience with order management, customer disputes, support, and platform policies.

No	Question	Answer / Finding
1	How easy was it for you to register and start selling on Daraz?	The registration process is relatively simple, and a seller can start listing products without a very complicated procedure.
2	What are the biggest challenges you face while selling on Daraz?	Competition, pricing pressure, managing orders during campaigns, and handling customer complaints are some of the major challenges.

3	How do you handle a customer who claims that a product is damaged, incorrect, or different from the description?	We usually check the order details and product condition and then follow Daraz's return or dispute process.
4	How satisfied are you with Daraz's support when there is a dispute between a seller and customer?	Support is available, but complicated disputes can sometimes take time to resolve. Faster communication and clearer decisions would be helpful.
5	Do Daraz's policies and policy changes reach sellers clearly?	Important policies are communicated through the seller platform, but during major campaigns there can be a lot of information to follow
6	What difficulties do you face during major campaigns such as 11.11?	Order volume increases significantly, so managing inventory, packing, responding to customers, and meeting deadlines becomes more difficult.
7	Do you think the seller rating system is fair?	The rating system is useful, but sellers may feel that some ratings are based on issues outside their direct control, such as delivery problems.

Table 3.4: Seller interview

Key Finding:

The seller perspective indicates that **Daraz provides a relatively accessible selling platform,** but sellers face challenges related to **competition, campaign-period workload, customer disputes, and communication with platform support.** This supports the report's existing finding that seller communication and dispute handling are important areas for improvement.

3.2.3.4 Customer Interview — Daraz Customer (Sifat Bhuyan)

- **Interviewee:** Daraz Customer (Sifat Bhuyan)
- **Platform:** Daraz Bangladesh
- **Purpose:** To understand the customer's experience from product selection to delivery, return, refund, and support.

No	Question	Answer / Finding
1	How often do you use Daraz?	I use Daraz occasionally or regularly, especially when there are discounts or campaigns.
2	What is the main reason you choose Daraz instead of buying from a physical store?	Product variety, convenience,discounts, and the ability to compare different sellers are the main reasons

3	Have you ever received a product that was different from what was shown online?	Yes, there can be situations where the actual product does not completely match the pictures or description
4	How satisfied are you with the return and refund process?	The return request is relatively easy to start, but waiting for the refund or getting clear status updates can sometimes be frustrating.
5	Have you ever experienced a delivery delay?	Yes. Delivery is usually completed within the estimated time, but delays can happen, particularly during busy periods.
6	What would you do if the app showed "delivery attempt failed" even though you were available?	I would contact customer support and ask them to check the delivery status and arrange another delivery attempt.
7	How would you rate Daraz customer support?	Support is useful for common problems, but more complicated issues may require additional followup.

Table 3.5: Customer interview

Key Finding

The customer interview highlights that the **normal shopping process is convenient,** but trust becomes weaker when problems occur. The major concerns are **product authenticity, delivery delays, refund visibility, and customer-support resolution,** which closely match the problems identified through the report's review analysis.

3.2.3.5 Delivery Partner Interview — Pathao Delivery Agent (Rakib)

Interviewee: Pathao Delivery Agent - Rakib
Role: Delivery Rider
Purpose: To understand the last-mile delivery process and the problems faced by delivery partners when delivering Daraz orders.

No.	Question	Answer / Finding

1	What are the most common problems you face while delivering e-commerce parcels?	Incorrect or incomplete addresses, customers not answering calls, traffic, and difficulty finding some locations are common problems.
2	What happens when you cannot reach the customer?	We normally try to contact the customer and may make another delivery attempt according to the delivery process.
3	What information do you receive about the customer and delivery?	We receive the necessary delivery information, including the customer's address and contact information required to complete the delivery.
4	How do you record a failed delivery attempt?	The delivery status is updated according to the reason the parcel could not be delivered, such as the customer being unavailable or the address being difficult to locate.
5	What happens if a customer says they were available but the delivery was marked as failed?	The delivery record and communication with the customer may need to be checked to understand what happened and determine whether another attempt is required.
6	What makes delivery difficult during major online campaigns?	The number of parcels increases significantly, while traffic and time pressure remain major challenges.
7	How useful is GPS or location information during delivery?	Location information is helpful for finding customers, but addresses in some areas may still be difficult to identify accurately.
8	What improvement would make delivery easier?	More accurate addresses, better location information, clearer customer instructions, and better communication between the marketplace and delivery partner would help.

Table 3.6: Delivery Partner Interview

Key Finding

The delivery-agent perspective shows that last-mile delivery depends heavily on accurate customer information, communication, location data, and coordination between the marketplace and courier . Failed deliveries may therefore result from several factors,not only customer

unavailability . This supports the report's observation that delivery-status accuracy and proof of delivery are areas where the existing system could be improved .

3.2.4 Questionnaires

A questionnaire is a structured set of questions used to collect information from a group of people in a consistent way. For this report , three separate questionnaires were designed : one aimed at Daraz customers, one aimed at Daraz sellers, and aimed at general app users. These questionnaires were designed to cover the five problems areas identified in chapter 1 , and to guide the review based content analysis presented later in section 3.3.

Questionnaire for Customers (Multiple Choice and Yes/No)

1. How often do you order products from Daraz ? (Daily/Weekly/Monthly/Rarely)
2. Have you ever received a product that was different from what was shown on the app ? (Yes or No)
3. How satisfied are you with the refund process on Daraz ? (Very Satisfied / Satisfied / Neutral / Unsatisfied / Very Unsatisfied)
4. Have you ever experienced a delivery delay beyond the estimated date ?(Yes /No)
5. Has your Daraz account ever been accessed without your permission ?(Yes/No)
6. How would you rate the customer support response time ?(Fast/Average/Slow/Very Slow)
7. Which payment method do you use most often? (Cash on Delivery / Mobile Financial Service / Card or Online Banking)

Questionnaire for Sellers (Ranking Scale, 1 = Lowest, 5 = Highest)

8. Rank your satisfaction with Daraz's seller-support tools. (1 to 5)
9. Rank how clearly Daraz communicates policy changes to sellers. (1 to 5)
10. Rank how fair you feel the seller-rating system is. (1 to 5)
11. Rank your satisfaction with how disputes with buyers are resolved. (1 to 5)
12. Rank your overall experience selling on Daraz. (1 to 5)

Questionnaire for App Users (Fill in the Blank and Multiple Choice)

13. I have used the Daraz app for _____ (less than 6 months / 6 months to 1 year / more than 1 year).
14. The feature I use most often on the Daraz app is _____.
15. My last order from Daraz arrived within _____ days.
16. What is your primary reason for using Daraz? (Price / Product variety / Convenience / Promotions and discounts)
17. Which device do you primarily use to shop on Daraz? (Mobile App / Website / Both)

18. How would you describe the pricing on Daraz compared to physical shops? (Much lower / Slightly lower / About the same / Higher)

These questionnaires were used to guide a structured content analysis of the large number of public customer reviews and comments already available for Daraz Bangladesh on platforms such as Trustpilot, SmartCustomer, Google Play Store, and Reviews.io. Rather than collecting brand-new responses, each publicly available review was read and matched against the closest related question above, and grouped into the five problem categories identified in Chapter 1. The results of this content analysis are presented as charts in Section 3.3.

3.3 Information Representation / Analysis

Once information has been gathered , it needs to be organized and presented in a way that is easy to understand . Charts and graphs are especially useful for this, because they can show patterns and proportions at a glance ,in a way that would be much harder to see from plain text alone . This section presents the results of the review-based content analysis described above ,using charts built from the patterns found across collected customer reviews.

3.3.1 Charts and Graphs

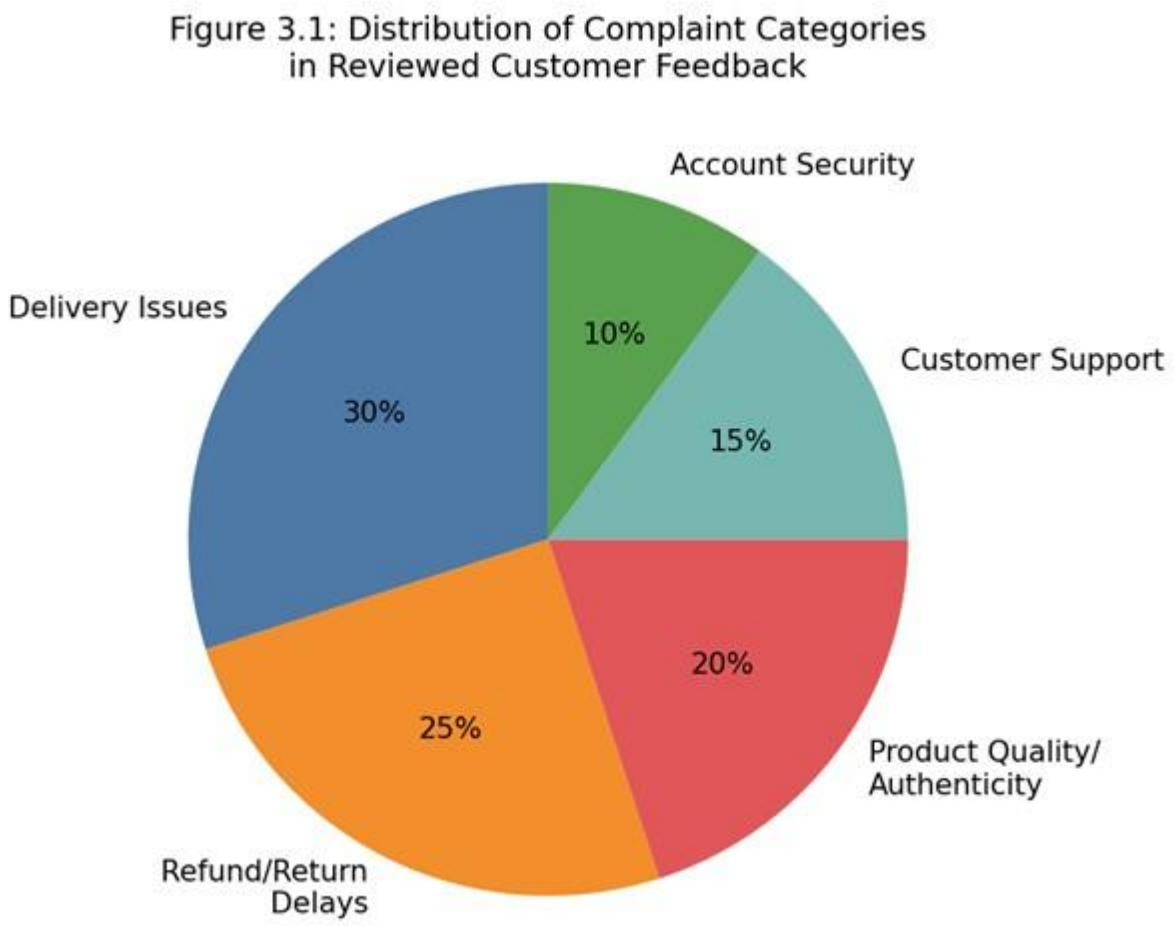


Figure 3.1: Distribution of complaint categories found across the customer reviews studied for this report

As Figure 3.1 shows, delivery-related issues make up the largest single share of complaints found in the reviews studied, followed closely by refund and return delays. Product authenticity and quality concerns form the next largest group, followed by customer support issues and, lastly, account-security related complaints. This pattern lines up closely with the five problems identified in Chapter 1, and confirms that these are not rare, one-off complaints, but issues that appear repeatedly across many different customers and review platforms.

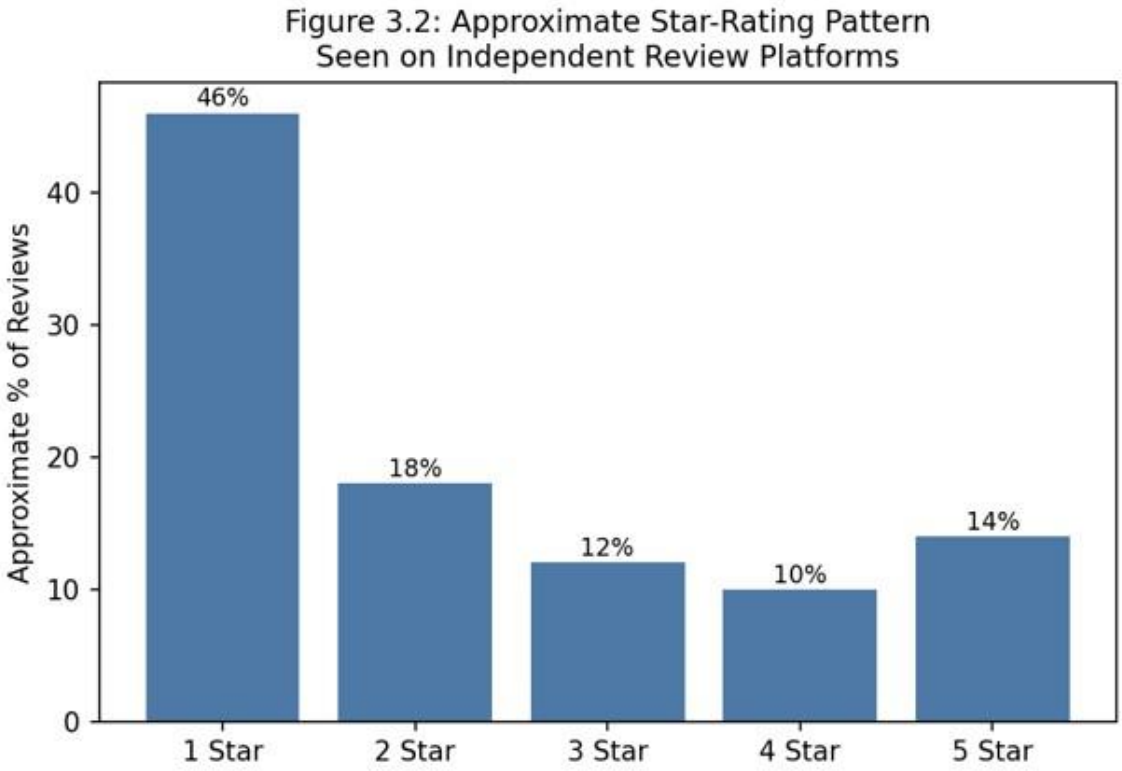


Figure 3.2: Pattern of star ratings seen on independent review platforms for Daraz Bangladesh.

Figure 3.2 shows an approximate pattern commonly seen on independent review platforms, where a large share of reviewers leave a very low, one-star rating, alongside a smaller but steady group of very satisfied, five-star reviewers. This kind of pattern is fairly common for large e-commerce platforms, since customers who had a smooth, ordinary experience are less likely to leave any review at all, while customers who had a serious problem are more likely to make the effort to write one. This means the star-rating pattern should be read as a picture of where problems tend to cluster, rather than as an exact measure of overall customer satisfaction.

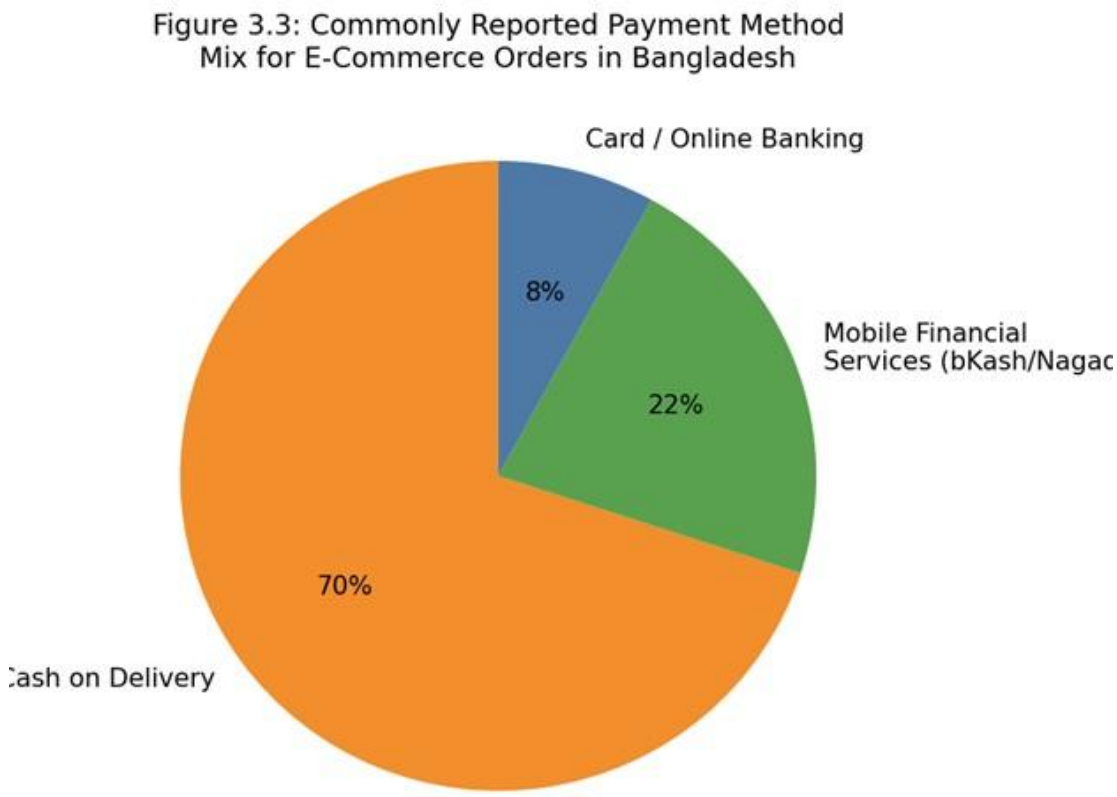


Figure 3.3: Commonly reported mix of payment methods used for online orders in Bangladesh

Figure 3.3 shows that Cash on Delivery remains the most commonly used payment method for online orders in Bangladesh by a wide margin, with mobile financial services such as bKash and Nagad forming the second largest group, and card or online-banking payments making up the smallest share. This pattern is an important context for the refund-process problem discussed earlier, since a return on a Cash on Delivery order means refunding real cash rather than simply reversing a digital transaction, which can make the refund process feel slower and more stressful for the customer.

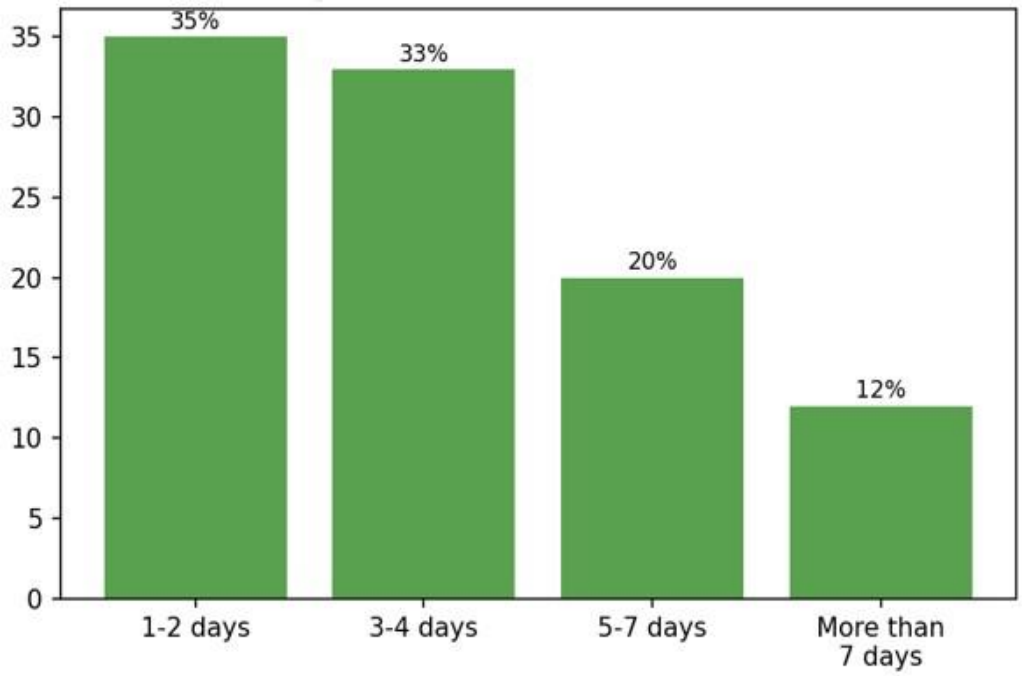


Figure 3.4: Commonly reported delivery time ranges mentioned in customer reviews.

Figure 3.4 groups delivery-time mentions found in the reviews into simple ranges. A little over two-thirds of mentions fall within a one-to-four-day range, which lines up with Daraz's publicly reported progress in shrinking delivery times over the past several years. Still, a noticeable share of mentions, around one in three, describe deliveries taking five days or longer, and it is this slower group that tends to overlap most closely with the delivery complaints discussed in Chapter 1 and Chapter 2, including disputed "failed delivery" statuses.

3.3.1.1 Reading These Charts Together

Looking at all four charts side by side tells a more complete story than any single chart on its own. Because Cash on Delivery is used for the large majority of orders (Figure 3.3),and because refund and return delays are the largest complaint category (Figure 3.1),it is reasonable to conclude that Cash on Delivery orders are disproportionately affected by the refund problems described in chapter 1 and chapter 2. Similarly, the heavy skew toward one-star reviews in Figure 3.2 does not mean most customers are unhappy overall;rather,it shows that unhappy customers are far more motivated to leave a review than satisfied ones,which is an important detail to keep in mind when interpreting any public review data ,including the analysis in this report.

3.3.2 DFD of Existing System

A Data Flow Diagram, or DFD,Is sometimes called a “bubble chart”. It is used to show how data moves through a system,how that data is transformed at each step along the way. A DFD does

not show every small technical detail of a system; it focuses on the main processes, the data stores they use, and the outside entities they connect to,such as customers,sellers, and delivery partners.

The diagram below shows the existing order-to-delivery workflow of Daraz Bangladesh, based on the order process observed and described in Section 3.2. . It begins when a customer searches for and selects a product, and ends with the order being delivered, with a separate path shown for return and refund requests.

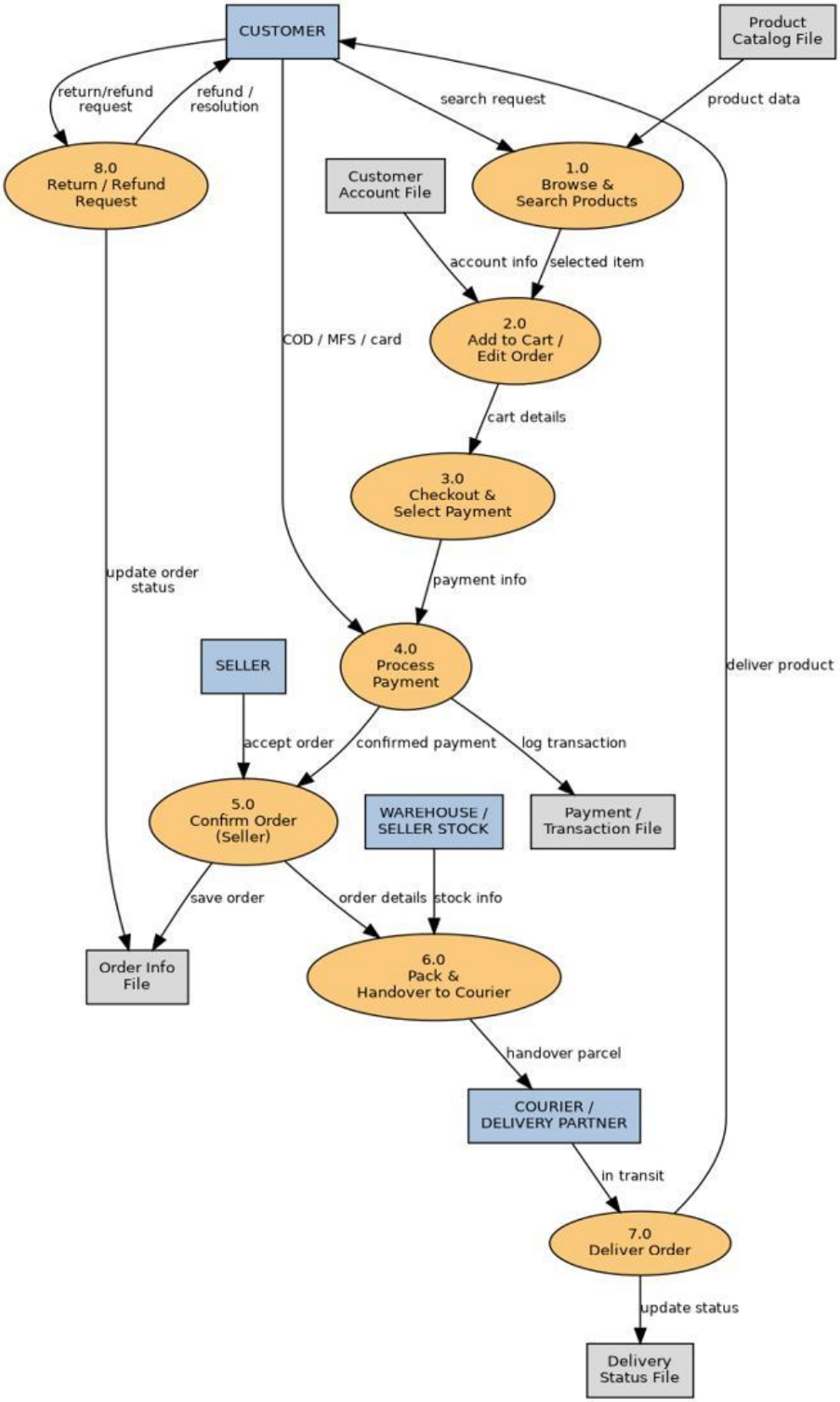


Figure 3.5: Data Flow Diagram (DFD) of the existing Daraz Bangladesh order and delivery system

The process begins with the Customer entity sending a search request to Process 1.0, Browse and Search Products, which pulls information from the Product Catalog File. Once a product is selected, Process 2.0, Add to Cart and Edit Order, uses information from the Customer Account File to prepare the order. The order then moves to Process 3.0, Checkout and Select Payment, where the customer chooses between Cash on Delivery, a mobile financial service, or a card payment.

Process 4.0, Process Payment, records the transaction in the Payment and Transaction File and passes the confirmed payment to Process 5.0, Confirm Order, where the Seller entity accepts the order. This step saves the order details into the Order Info File. Next, Process 6.0, Pack and Handover to Courier, uses stock information from the Warehouse or Seller Stock entity to prepare the parcel, which is then handed to the Courier or Delivery Partner entity.

Process 7.0, Deliver Order, updates the Delivery Status File once the product reaches the customer. A separate path, Process 8.0, Return and Refund Request, allows the customer to start a return at any point after delivery, which updates the Order Info File and eventually leads to a refund or another resolution being sent back to the customer. This diagram makes it easier to see exactly where in the process each of the five problems identified in Chapter 1 tends to occur. Product authenticity issues trace back to the Product Catalog File and Process 5.0, since this is where seller and product information first enters the system. . Refund delays connect to the weak feedback loop around Process 8.0. Account security issues relate to Process 4.0 and the Customer Account File. Delivery problems cluster around Process 6.0 and 7.0. Customer support issues cut across the whole diagram, since support staff need accurate information from every file shown here to help a customer properly.

3.4 Cost-Benefit Analysis

A cost-benefit analysis compares the total cost of building and running a proposed improvement against the total benefit or saving it is expected to bring. This helps decide whether a proposed change is financially worth doing. For Daraz Bangladesh, the proposed improvements from Chapter 2 mainly involve adding new checks, tracking tools, and communication features on top of the existing system, rather than replacing the whole platform, which keeps the overall cost more manageable.

3.4.1 Estimated Cost of Proposed Improvements

Before looking at specific figures, it helps to separate costs into a few simple categories. Tangible costs are the ones that are easy to measure directly, such as paying developers to build a new feature or paying for extra server capacity.Intangible costs are harder to put an exact number on, such as the short-term inconvenience customers might feel while a new verification step is being rolled out.Similarly, some costs are fixed, meaning they do not change much with the number of orders, such as the one-time cost of building a fraud-detection tool, while other costs are variable, meaning they grow with usage, such as the cost of sending extra SMS verification messages as more customers use the platform.

The table below gives a simple, estimated breakdown of the type of costs involved in building the improvements recommended in Chapter 2, such as stronger seller verification, better return tracking, extra account-security steps, improved delivery-proof tools, and better support routing. These figures are illustrative estimates used for planning purposes, based on typical costs for this type of feature work.

Cost Type	**Item**	**Estimated Cost (USD)**
Technology / Development Cost	Seller verification and rating system	$10,000
	Fraud detection and login-security upgrade	$12,000
	Return and refund status tracking tool	$5,000
	Delivery proof-of-attempt feature	$3,500
Personnel Cost	Additional support staff and training	$7,000
Operating Cost	Awareness campaigns on fraud and fake pages	$2,500
Total Estimated Cost		**$40,000**

Table 3.7: Estimated cost breakdown for the proposed system improvements.

3.4.2 Break-Even Point Analysis

The break-even point is the point at which the total benefit gained from a change becomes equal to its total cost, meaning the project neither gains nor loses money at that point. Any activity beyond the break-even point can be considered a net gain. For this analysis, we treat each successfully completed order that avoids a costly problem, such as a fraud case, a bad refund experience, or a failed delivery, as one "unit" of benefit, worth an estimated $13 in saved cost and retained customer value, against an estimated variable cost of $9.5 per unit to maintain the new checks and features.
Using a fixed cost of $40,000 and the per-unit numbers above, the break-even point can be calculated using the standard formula:

Break-Even Point (in units) = Fixed Cost ÷ (Price per unit − Variable Cost per unit) = 40,000 ÷ (13 − 9.5) = 40,000 ÷ 3.5 ≈ 11,429 units.

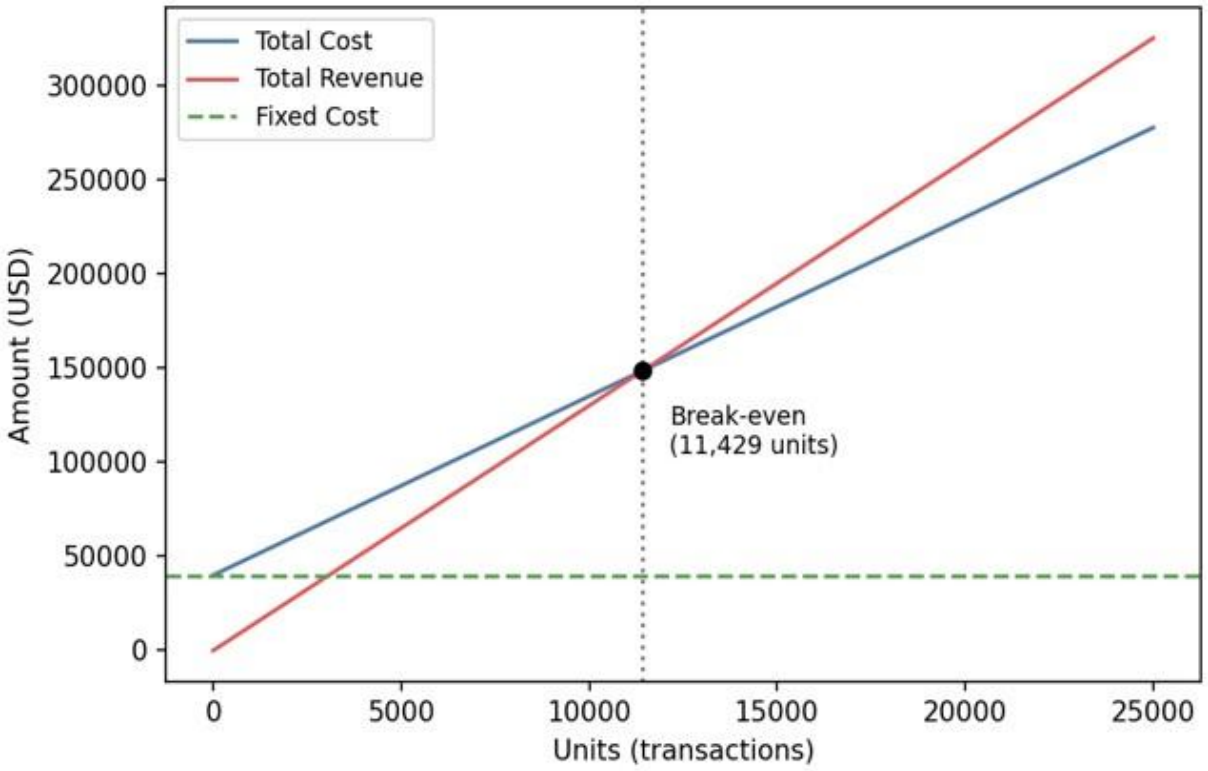


Figure 3.6: Break-even analysis for the proposed system improvements.

This means that once the improved system successfully handles around 11,429 problem-avoiding order interactions, such as catching a fake product before it ships, resolving a refund quickly, or preventing a fraudulent login, the cost of building and running the new features will have paid for itself. Given that Daraz Bangladesh handles a very large number of orders every day, reaching this break-even point is realistic within a fairly short time after launch, which supports the economic feasibility conclusion already reached in Chapter 2.

3.5 Detailed Feasibility Analysis

3.5.1 Root Cause Analysis of Identified Challenges

This section looks one level deeper into each of the five problems identified earlier in this report, to understand not just what is going wrong, but why it happens, and what effect it has on both the system's performance and the way customers behave.

1. Product Authenticity and Quality Control

Root Cause: Daraz's marketplace model relies on thousands of independent sellers, and the process for checking a seller's identity and product quality before they are allowed to sell is not strict enough to catch every case of a fake or low-quality product.

Performance Impact: Fake or low-quality products lead to more returns, more refund requests, and more customer-support tickets, which adds extra workload across the whole system.

Behavioral Impact: Customers who receive a bad product become more cautious, often avoiding certain categories or sellers entirely, and are more likely to share their bad experience publicly, which can discourage new customers from trusting the platform

2. Refund and Return Process

Root Cause: The return and refund process does not give customers clear, trackable status updates, and does not always explain why a request was rejected or cancelled.

Performance Impact: Unclear refund status leads to a higher number of repeated customersupport contacts, since customers reach out again and again to check on the same request.

Behavioral Impact: Customers who feel unsure about getting their money back become hesitant to buy higher-priced items, and may reduce how often they use Cash on Delivery, even though it is normally their preferred payment method.

3. Account Security and Fraud

Root Cause: The current system does not have strong enough tools to detect unusual account activity early, and there is no simple, fast way for customers to freeze their account the moment they suspect a problem.

Performance Impact: Successful account hacking or fraud cases require manual investigation and customer-support time to resolve, and in some cases cannot be fully reversed, such as a gift card that has already been claimed.

Behavioral Impact: Customers who hear about hacking or fake-website cases,even if it did not happen to them personally, may become more cautious about saving payment details or trusting promotional links related to Daraz.

4. Delivery Service

Root Cause: Once a parcel is handed over to a courier partner, Daraz's visibility into the exact reason for a delivery delay or failed attempt becomes limited, and there is no simple way for a customer to challenge or clarify a delivery status they disagree with

Performance Impact: Delivery delays and disputed "failed delivery" statuses lead to more redelivery attempts, more customer-support contacts, and in some cases, cancelled orders that need to be reversed.

Behavioral Impact: Customers who experience a delivery problem may reduce trust in the delivery time estimates shown at checkout, and may become less willing to order time-sensitive items.

5. Customer Support and Seller Communication

Root Cause: Customer support relies heavily on an automated chatbot for the first stage of contact, and there is no strict rule requiring a chat to stay open until the customer's issue is actually resolved. Seller communication speed is also not closely tracked or enforced.

Performance Impact: Support conversations that end without solving the real issue often lead to the customer contacting support again, effectively duplicating the same work.

Behavioral Impact: Customers who feel unheard by support are more likely to leave a negative public review, even if the original problem, such as a late delivery, was relatively minor and could have been resolved quickly with better communication.

3.5.2 Root Cause Summary Table

The table below brings together the root cause, performance impact, and behavioral impact for all five problems in one place, for a quick side-by-side comparison.

Problem	Root Cause	Performance Impact	Behavioral Impact
Product Authenticity	Weak seller and product checks before listing	More returns and support tickets	Customers avoid certain sellers/categories
Refund and Return	No clear, trackable refund status	Repeated support contacts for same request	Hesitation toward higherpriced orders
Account Security	Weak early fraud detection and account freeze options	Manual investigation needed per case	Caution around saved payment details
Delivery Service	Limited visibility once handed to courier	More re-delivery attempts, cancellations	Lower trust in delivery time estimates
Customer Support	Heavy reliance on chatbot, no closure rule	Duplicate support contacts	More negative public reviews

Table 3.8: Root cause, performance impact, and behavioral impact across all five problems.

This side-by-side view makes it clear that most root causes across all five problems point back to the same general gap: the current system is better at handling the ordinary, problem-free order than

at handling the exceptions, such as a fake product, a disputed delivery, or a hacked account. This suggests that Daraz Bangladesh's overall business model and technology base are sound, but the "exception-handling" layer of the system, meaning the tools and processes that activate once something goes wrong, is where the most valuable improvements can be made.

It is worth being clear about the limits of the approach used in this chapter, so that the findings are read with the right amount of confidence. First, the review-based content analysis in Section 3.3 relies on public reviews, which tend to over-represent very unhappy or very happy customers, while customers who had an average, uneventful experience are less likely to leave a review at all.This means the charts in this chapter are useful for spotting where problems cluster, but should not be read as an exact measurement of what percentage of all Daraz customers are satisfied overall.

Second, the helpline interview in Section 3.2.3 reflects a single call to a single support agent, so it captures one useful, real data point rather than a broad, statistically representative sample of how every agent handles every situation. Third, direct access to Daraz Bangladesh's senior management and internal operational data was not available for this project, so some of the analysis, particularly around internal costs and root causes, is built on reasonable estimation and publicly available information rather than confirmed internal figures

None of these limitations make the analysis in this chapter less useful. Large, well-known patterns, such as the repeated complaints about delivery, refunds, and account security, show up consistently across many independent sources, which gives good confidence that these are real and recurring issues rather than a one-time coincidence. The limitations simply mean that specific numbers, such as the exact break-even point or the precise percentage of complaints in each category, should be treated as reasonable estimates for planning purposes, not as final, audited figures.

3.6 Conclusion

This chapter has gone through the full information gathering and analysis processes for Daraz Bangladesh’s existing system. Using a review of literature, procedures, and forms, close observation of the platform itself, a real helpline call combined with the company’s own published customer-service materials and a structured content analysis of public reviews, a clear and consistent picture has formed around the five problems first identified in chapter 1.
The Data Flow Diagram helped show exactly where in the order-to-delivery process each problem tends to occur, while the cost-benefit and break-even analysis showed that the proposed improvements are realistic to fund and are likely to pay for themselves within a reasonable amount of activity. The root cause analysis then dug one level deeper, showing that most of the problems come from gaps in checking, tracking,and communication, rather than from anything fundamentally broken in Daraz’s overall business model.

With the detailed analysis complete ,the project is ready to move into chapter 4, where a redesigned candidate system will be proposed. This will include an updated Data Flow Diagram showing the improved process, a database design using an Entity-Relationship diagram and sample form designs that directly address the problems and root causes identified throughout this chapter.

Chapter 4

Design

4.1 Introduction

System design is the stage where everything learned during analysis is turned into an actual plan for how the new or improved system will work. While Chapter 3 focused on understanding the

existing Daraz Bangladesh system and the problems inside it, this chapter focuses on building a clear, practical design that directly solves those problems.

Good system design usually involves a few key activities: deciding on the overall structure of the system, defining how data will move between its different parts, designing the underlying database, and designing the actual screens and forms that users will interact with. A system design should not just look good on paper; it needs to be realistic to build, easy for users to understand, and strong enough to handle the scale at which Daraz Bangladesh operates.

This chapter follows that structure. Section 4.2 proposes two candidate systems, compares them, and selects the stronger option, along with a detailed Data Flow Diagram (DFD) for the chosen design.

Section 4.3 covers the database design, including an Entity-Relationship (ER) diagram and a data dictionary describing each table. Section 4.4 presents the design of key new forms and screens needed to support the proposed system. Section 4.5 closes the chapter with a short summary.

4.2 DFD of the Proposed Candidate System

Before settling on a single design, it is good practice to consider more than one possible way to solve the identified problems, and to compare them before choosing.

4.2.1 Candidate System 1: Comprehensive Automated Trust and Security System

This candidate system takes a fuller, more automated approach to solving Daraz's problems. It introduces a Seller Verification and Product Authenticity Check process that requires new sellers to submit identity and business documents before their shop goes live, and continuously updates a visible Trust Score for each seller based on order history, complaint rate, and delivery performance.

For account security, this system adds a Fraud Detection and OTP Verification step at checkout, which uses a risk-scoring engine to flag unusual account activity, such as a login from a new device followed immediately by a high-value order, and requires a one-time SMS code before the order can proceed in risky cases. For account security, this system adds a Fraud Detection and OTP Verification step at checkout, which uses a risk-scoring engine to flag unusual account activity, such as a login from a new device followed immediately by a high-value order, and requires a onetime SMS code before the order can proceed in risky cases. Finally, a Support Ticket and Escalation process guarantees that any issue not resolved by the chatbot within a set time is automatically handed to a human agent with a tracked service-level deadline.

4.2.2 Candidate System 2: Incremental, Lower-Cost System

This candidate system aims to solve the same five problems, but with a lighter, lower-cost approach that could be rolled out faster. Instead of a full document-based seller verification process, sellers are grouped into simple trust tiers (New, Standard, Verified) based only on order history and complaint counts that Daraz already collects, without requiring new document uploads.

For account security, instead of a full risk-scoring engine, this system uses simpler rule-based checks, such as automatically requiring an SMS OTP any time a password is changed or a new delivery address is added, without a continuous behind-the-scenes scoring system. For account security, instead of a full risk-scoring engine, this system uses simpler rule-based checks, such as automatically requiring an SMS OTP any time a password is changed or a new delivery address is added, without a continuous behind-the-scenes scoring system. For delivery, riders confirm delivery using a shared OTP code only, without a photo requirement, which is faster for the rider but offers slightly weaker proof in disputed cases. For returns, customers get a simple SMS update at each stage of the return, rather than a fully trackable in-app status screen. Support tickets are still escalated to a human agent when unresolved, but without a strict, tracked time deadline attached to each case.

4.2.3 Comparison of Candidate Systems

The table below gives a simple, qualitative comparison of the two candidate systems across the criteria that matter most for this project: how strong the fraud and authenticity protection is, how good the customer experience is, and how much it costs and how long it takes to build.

Evaluation Criteria	**Candidate System 1**	**Candidate System 2**
Seller verification strength	Very good (document-based)	Satisfactory (history-based only)
Fraud detection strength	Very good (risk-scoring engine)	Good (rule-based checks)

Delivery dispute protection	Very good (code + photo)	Satisfactory (code only)
Refund transparency	Very good (full tracking)	Good (SMS updates only)
Support resolution speed	Very good (tracked SLA)	Good (no strict SLA)
Estimated build cost	Higher	Lower
Estimated build time	Longer (phased rollout needed)	Shorter (faster to launch)

Table 4.1: Qualitative comparison of the two candidate systems

To make the comparison more concrete, a simple weighted scoring exercise was also carried out. Each criterion was given a weight based on how important it is to solving the problems identified earlier in this report, and each candidate system was scored from 1 to 5 against that criterion.

Criteria	Weight	System 1 Rating	System 1 Score	System 2 Rating	System 2 Score
Seller verification	5	5	25	3	15
Fraud detection	5	5	25	4	20
Delivery dispute protection	4	5	20	3	12
Refund transparency	4	5	20	4	16
Support resolution speed	3	5	15	4	12
Cost-effectiveness	3	3	9	5	15
Total			**114**		**90**

Table 4.2: Weighted evaluation matrix for the two candidate systems.

4.2.4 Selecting the Best Candidate System

Based on the weighted evaluation, candidate system 1 scores noticeably higher overall, mainly because it offers stronger protection in exactly the areas that were shown, in chapter 1 through 3, to matter most: seller verification, fraud detection,and delivery-dispute protection. Candidate System 2 remains a reasonable lower-cost fallback option if Daraz Bangladesh wants to roll out improvements in smaller, faster stages, but Candidate System 1 is recommended as the primary design to move forward with, since it more fully addresses the root causes identified earlier in this report. The remainder of this chapter is based on Candidate System 1.

4.2.5 DFD of the Selected Candidate System

The diagram below shows the proposed candidate system's order-to-delivery workflow, highlighting the new processes added to address the problems identified earlier in this report.

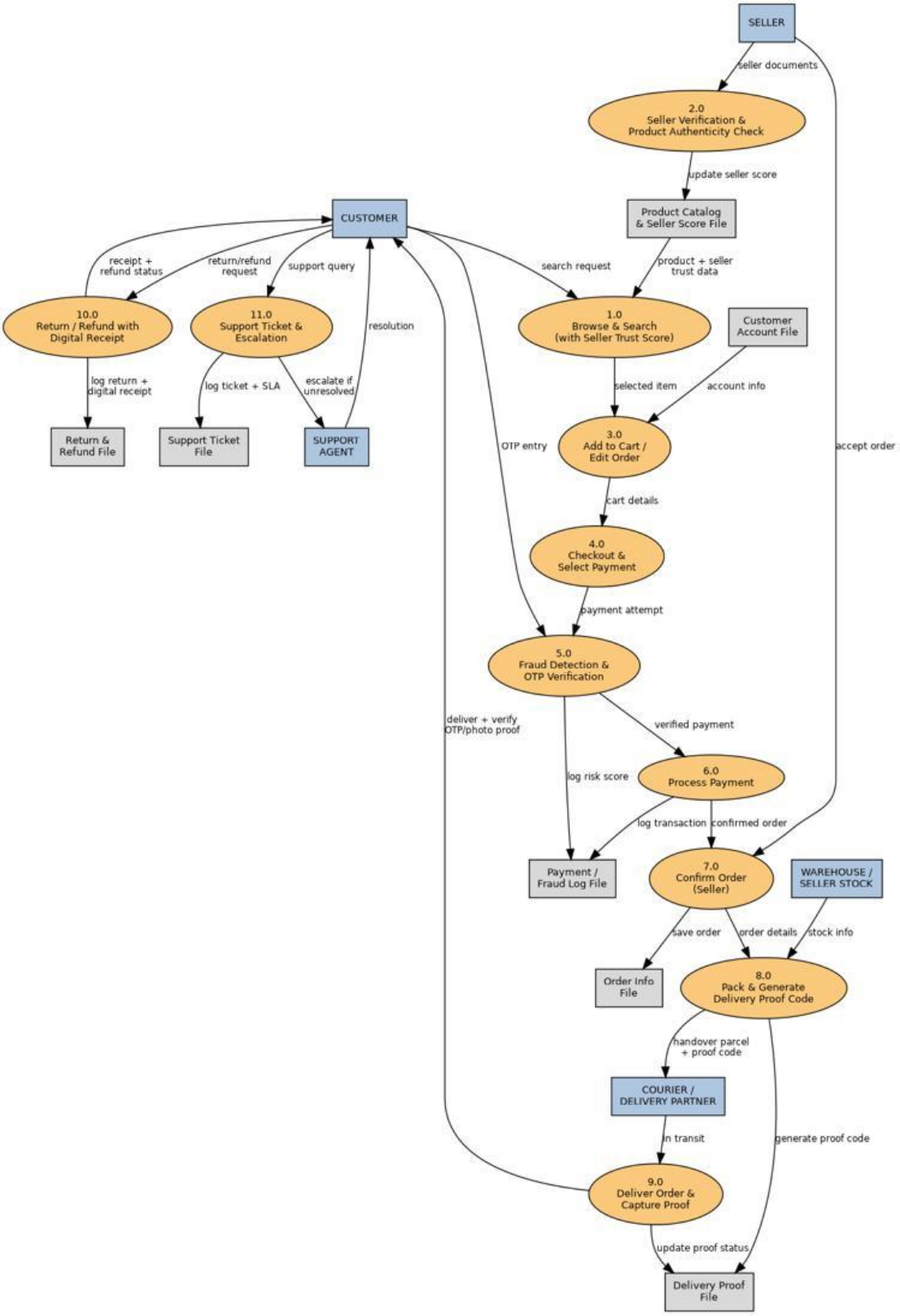


Figure 4.1: Data Flow Diagram (DFD) of the proposed candidate system (Candidate System 1).

The process begins the same way as the existing system, with the Customer browsing and searching for products, but Process 1.0 now also pulls a Seller Trust Score from the Product Catalog and Seller Score File, so customers can see this information before buying. Process 2.0, Seller Verification and Product Authenticity Check, is a new process where the Seller submits identity and business documents, which updates their trust score in the catalog file.

At checkout, Process 5.0, Fraud Detection and OTP Verification, is a new step that checks the payment attempt for risk factors and, when needed, asks the Customer to confirm a one-time code before the order proceeds. This risk score is logged in the Payment and Fraud Log File. Once payment is verified, Process 6.0 processes the payment as before, and Process 7.0 lets the Seller confirm the order.

Process 8.0, Pack and Generate Delivery Proof Code is another new addition: instead of simply heading the parcel to the courier, the system generates a proof code tied to that specific order. Process 9.0, Deliver Order and capture proof requires the rider to collect this code (and optionally a photo) directly from the customer at the point of delivery, which is logged in the new Delivery Proof File.

Process 10.0, Return and Refund with Digital Receipt, immediately issues a digital receipt the moment a return is logged, and tracks the refund status in the Return and Refund File. Finally, Process 11.0, Support Ticket and Escalation, logs every support query in a Support Ticket File with an attached service-level deadline, and automatically routes the case to a human Support Agent if it is not resolved in time.

4.2.6 How the Proposed System Solves the Identified Problems

Problem (Chapter 1)	New Process in Proposed System	How It Helps
Product Authenticity and Quality Control	Process 2.0 — Seller Verification and Product Authenticity Check	Sellers are checked before selling, and a visible trust score warns customers about risky sellers before they buy.
Refund and Return Process	Process 10.0 — Return and Refund with Digital Receipt	Every return is immediately confirmed with a digital receipt, and refund status becomes trackable step-by-step.
Account Security and Fraud	Process 5.0 — Fraud Detection and OTP Verification	Risky logins or purchases require extra confirmation, catching fraud before an order completes.
Delivery Service	Process 8.0 & 9.0 — Delivery Proof Code and Capture	A delivery cannot be marked complete or failed without a proof code or logged attempt, reducing disputes.
Customer Support and Seller Communication	Process 11.0 — Support Ticket and Escalation	Unresolved chatbot conversations are automatically escalated to a human agent within a tracked time limit.

Table 4.3: Mapping of each identified problem to its corresponding new process in the proposed system

4.3 Database Design

A well-structured database is the backbone of any system design, since it is where all the information generated by the new process above needs to be stored, connected and retrieved efficiently. This section presents the objectives behind the proposed database, followed by an Entity-Relationship (ER) diagram and a data dictionary describing each table in detail.

4.3.1 Objectives

- Efficient Data Store all information related to customers, sellers, products, orders, payments, deliveries, return and support tickets in a structured, non-repetitive way.

- Support for New Trust and security Features: Provide dedicated fields for seller-trust scores, fraud risk scores, OTP verification status and delivery proof codes, so the new processes described in Section 4.2 have a proper place to store their data

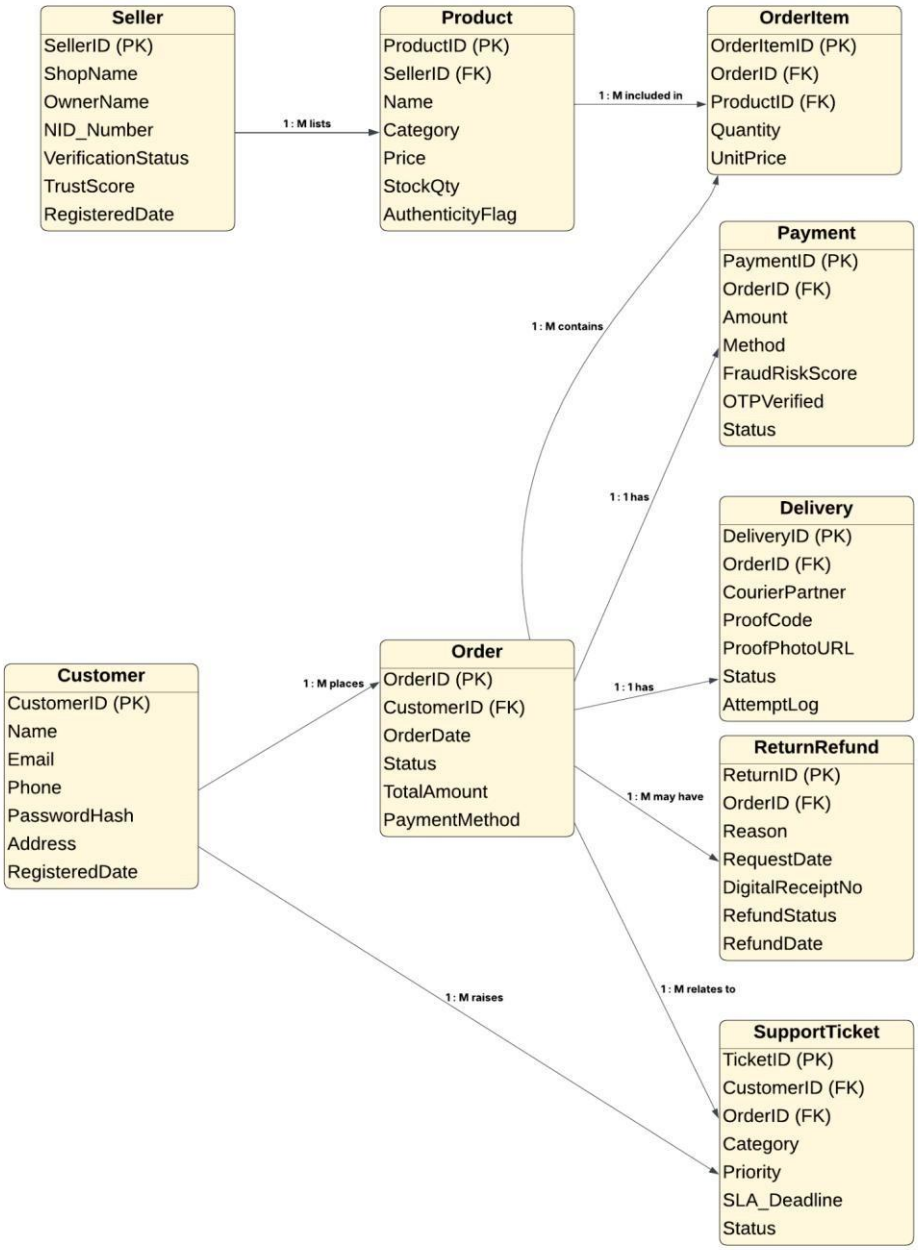
- Data integrity: Use primary and foreign keys to make sure that every order, payment, delivery, return, and support ticket is correctly linked back to the right customer, seller and product.

- Scalability: Design tables so that they can handle a very large number of daily orders and users without needing a major redesign as Daraz Bangladesh continues to grow.

- Support for Reporting: Make it possible to easily generate reports such as “average refund time” or "average delivery dispute rate per seller” which are useful for monitoring whether the proposed system is actually working as intended.

4.3.2 Entity Relationship (ER) Diagram

The Entity-Relationship diagram below shows the main tables (entities) in the proposed database, their key attributes, and how they connect to one another.



At the center of the design is the Order table, which connects to almost every other table: a Customer places one or more Orders, each Order contains one or more OrderItems drawn from the Product table, and each Order has exactly one Payment record and one Delivery record.An Order may also be linked to one or more ReturnRefund records if the customer requests a return, and to one or more SupportTicket records if the customer needs help. The Seller table connects to Product, since each product is listed by exactly one seller,and this is the table where the new seller Trust Score is stored.

4.3.3 Data Type and Size According to the ER Diagram

The tables below describe each entity from the ER diagram in more detail, listing every attribute along with its data type and size, similar to how a real database schema would be documented.

Customer Table

Attribute	Data Type	Size / Notes
CustomerID (Primary Key)	varchar	12
Name	varchar	50
Email	varchar	50
Phone	varchar	15
PasswordHash	varchar	128 (stored as a secure hash, never in plain text)
Address	varchar	150
RegisteredDate	date	—

Seller Table

Attribute	Data Type	Size / Notes
SellerID (Primary Key)	varchar	12
ShopName	varchar	50
OwnerName	varchar	50
NID_Number	varchar	20 (encrypted at rest)
VerificationStatus	varchar	15 (Pending / Verified / Rejected)
TrustScore	int	3 (0 to 100 scale)
RegisteredDate	date	—

Product Table

Attribute	Data Type	Size / Notes
ProductID (Primary Key)	varchar	12
SellerID (Foreign Key)	varchar	12
Name	varchar	80
Category	varchar	30
Price	decimal	10.2
StockQty	int	6
AuthenticityFlag	varchar	10 (Clear / Under Review / Flagged)

Order and OrderItem Tables

Attribute	Data Type	Size / Notes
OrderID (Primary Key)	varchar	14
CustomerID (Foreign Key)	varchar	12
OrderDate	datetime	—
Status	varchar	20 (Placed/ Confirmed/ Shipped / Delivered / Returned)
TotalAmount	decimal	10.2
PaymentMethod	varchar	15 (COD / MFS / Card)
OrderItemID (Primary Key)	varchar	14
OrderID (Foreign Key)	varchar	14
ProductID (Foreign Key)	varchar	12
Quantity	int	4
UnitPrice	decimal	10.2

Payment Table (with New Fraud-Detection Fields)

Attribute	Data Type	Size / Notes
PaymentID (Primary Key)	varchar	14
OrderID (Foreign Key)	varchar	14
Amount	decimal	10.2
Method	varchar	15
FraudRiskScore	int	3 (0 to 100 scale; new field)
OTPVerified	boolean	Yes/No (new field)
Status	varchar	15 (Pending / Completed / Flagged)

Delivery Table (with New Proof-of-Delivery Fields)

Attribute	Data Type	Size / Notes
DeliveryID (Primary Key)	varchar	14
OrderID (Foreign Key)	varchar	14
CourierPartner	varchar	20 (Pathao / Paperfly / REDX / eCourier)
ProofCode	varchar	6 (new field)
ProofPhotoURL	varchar	150 (new field, optional)
Status	varchar	20 (Out for Delivery / Delivered / Failed Attempt)

AttemptLog	text	Stores timestamped call/visit attempts (new field

ReturnRefund Table (with New Digital Receipt Field)

Attribute	Data Type	Size / Notes
ReturnID (Primary Key)	varchar	14
OrderID (Foreign Key)	varchar	14
Reason	varchar	100
RequestDate	datetime	—
DigitalReceiptNo	varchar	14 (new field, generated instantly)
RefundStatus	varchar	20 (Requested / Approved / Rejected / Completed)
RefundDate	date	Nullable until refund is completed

SupportTicket Table (with New SLA Field)

Attribute	Data Type	Size/Notes
TicketID (Primary Key)	varchar	14
CustomerID (Foreign Key)	varchar	12
OrderID (Foreign Key)	varchar	14 (nullable, not all tickets relate to an order)
Category	varchar	30 (Refund / Delivery / Account / Product / Other
Priority	varchar	10 (Low / Medium / High)
SLA_Deadline	datetime	New field; auto-set based on priority
Status	varchar	15 (Open / Escalated / Resolved)

4.4 Form Design

This section presents wireframe designs for the key new or modified forms needed to support the proposed system. Unlike the existing forms discussed in Chapter 3, which mostly already work well for ordinary orders, the forms below are new or meaningfully changed to directly support the new processes described in section 4.2.

4.4.1 Seller Verification Form

This form is completed by a new seller before their shop is allowed to go live. It supports process 2.0 from the proposed DFD, collecting identity and business information that feeds into the seller's Trust Score.

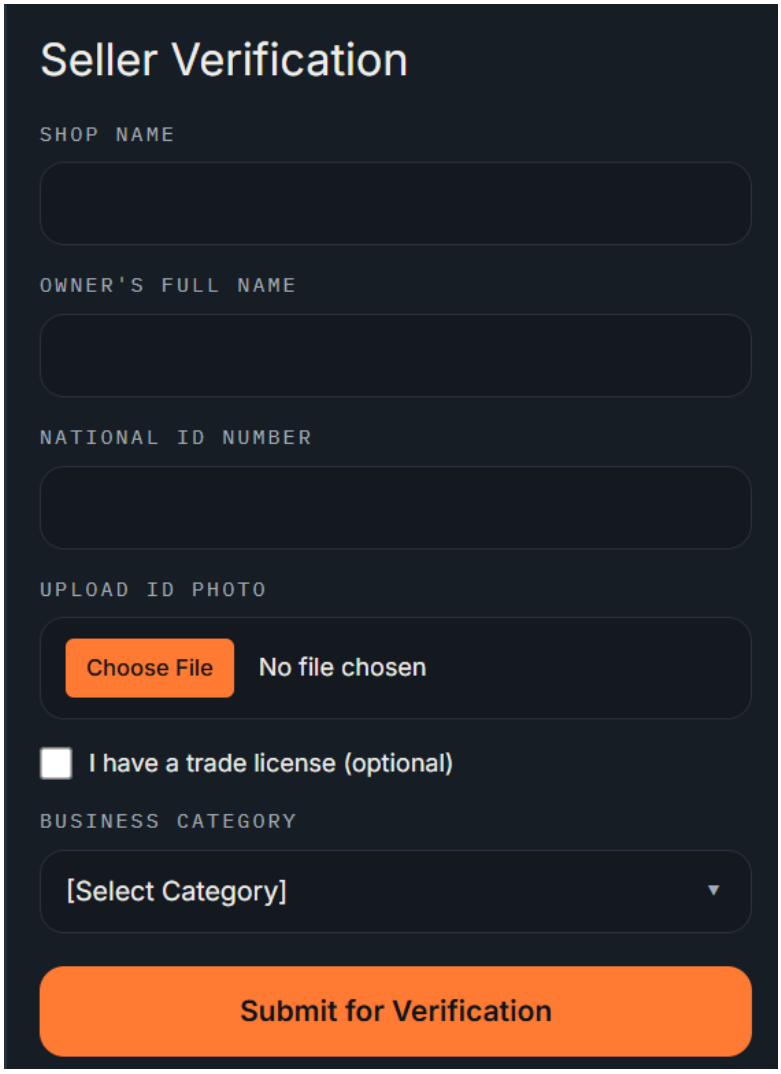


Figure 4.3: Wireframe design of the proposed Seller Verification form

This form asks for the shop name, the owner's full name and a National ID number, along with an uploaded photo of the ID itself. An optional trade license upload is available for sellers who have one, which can further increase their trust score once verified. A business category dropdown helps route the listing into the correct product category for later authenticity checks.

4.4.2 OTP / Account Security Verification Form

This form supports process 5.0, Fraud Detection and OTP Verification. It only appears when the system's risk-scoring engine flags an activity as unusual, such as a login from a new device followed by a high-value order.

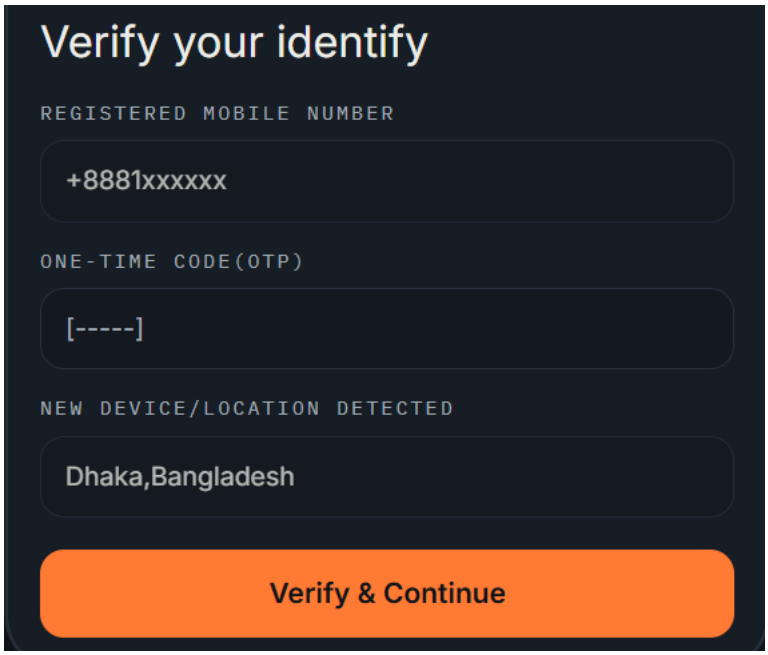


Figure 4.4: Wireframe design of the proposed OTP/Account security Verification form.

The form shows the customer's registered mobile number and asks them to enter a six-digit onetime code sent by SMS. It also clearly displays the location and date of the activity that triggered this extra check, so the customer can immediately recognize whether this was really them. A clearly visible "This wasn't me" option lets the customer instantly freeze their account if they did not request this login or purchase, directly addressing the account-security identified in Chapter 1.

4.4.3 Return / Refund Request Form

This form supports process 10.0, Return and Refund with Digital Receipt. It replaces the simpler existing return form discussed in Chapter 3 with one that immediately confirms the request with a trackable digital receipt.

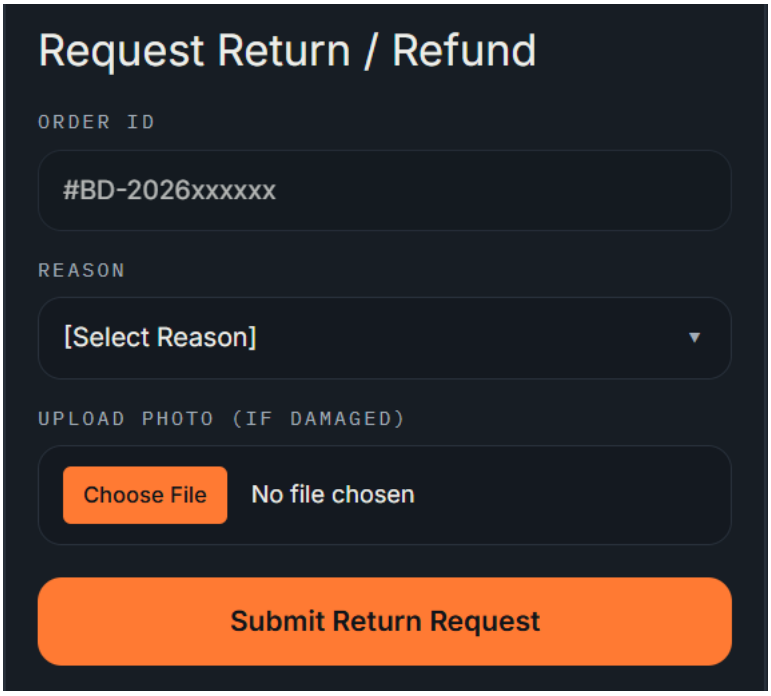


Figure 4.5: Wireframe design of the proposed Return / Refund Request form.

The customer selects the order, chooses a reason for the return from a dropdown list and can optionally upload a photo if the product arrived damaged. Once submitted, the system immediately generates a Digital Return Receipt with its own tracking ID, directly solving the problem identified in Chapter 1 and Chapter 2, where customers were previously left without any confirmation that their return had been received.

4.4.4 Delivery Proof Capture Form(Rider App)

This form supports process 9.0, Deliver Order and Capture Proof. It is used by the delivery rider's own app at the moment of delivery, not by the customer directly.

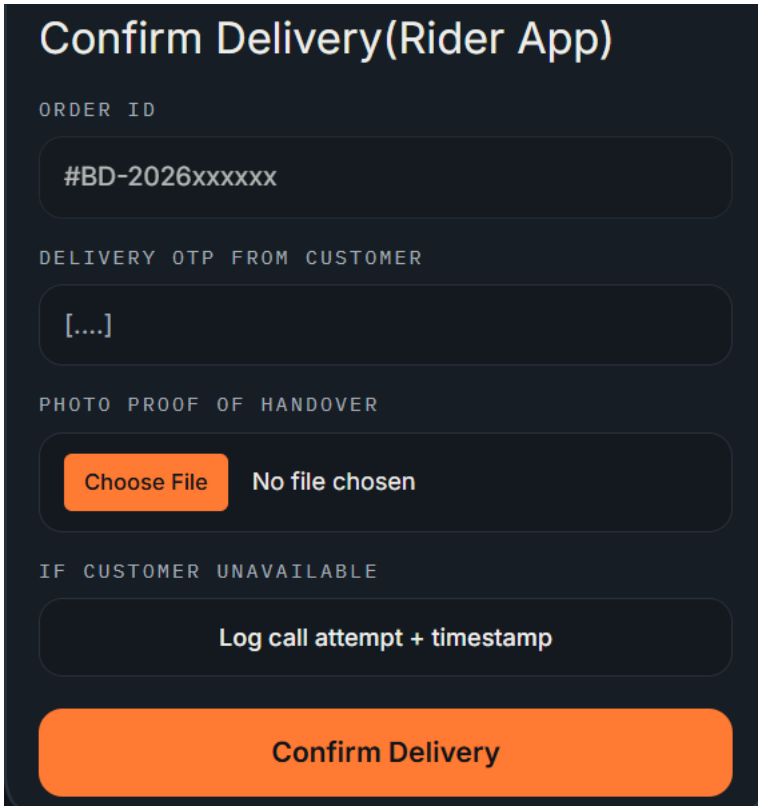


Figure 4.6: Wireframe design of the proposed Delivery proof Capture form of the Rider App.

At the point of delivery, the rider must enter the delivery OTP shown to the customer or capture a quick photo of the handover, before the order can be marked as delivered. If the customer is not available, the rider is required to log a call attempt along with a timestamp before the system will accept a "failed delivery" status. This directly addresses the delivery-dispute problem identified in Chapter 1, where deliveries were sometimes marked as failed without any way for the customer to verify what actually happened.

4.5 Conclusion

This Chapter has taken the problems and root causes identified in Chapter 1 through 3 and turned them into an actual system design. Two candidate systems were proposed and compared using both a qualitative comparison and a weighted scoring matrix and Candidate system 1, the more comprehensive,automated option, was selected as the recommended design.

A detailed Data Flow Diagram showed exactly how the new processes, seller verification, fraud detection, delivery proof capture, digital return receipts and support ticket escalation, fit into the

existing order-to-delivery workflow. The database design, including the Entity-Relationship diagram and the full data dictionary, showed how this new information can be stored in a structured, connected way. Finally, four key form designs showed what these new processes would actually look like from the point of view of a seller, a customer and a delivery rider.

Together,these design elements form a complete and realistic answer to the problems first identified in Chapter 1 of this report. If implemented, this design would give Daraz Bangladesh stronger seller accountability, better fraud protection,clearer delivery accountability, faster and more transparent refunds and a more reliable customer support process, addressing the root causes identified throughout this report rather than only their surface symptoms.